

7. $x \in L, x' \in L$

$$\Rightarrow L = x \times x'$$

8. line $L = (a, b, c)^T, L' = (a', b', c')^T$

$$\Rightarrow L \times L' = (c' - c)(b, -a, 0)^T$$

$\therefore L, L'$ meet at infinity

9. point $(x_1, x_2, 0)^T$ is known as

ideal points, or point at infinity.

$l_{\infty} = (0, 0, 1)^T$ is line at infinity.

9.3 $l_{\infty} = \{ \text{ideal points} \}$

10 $\forall$ point $x \in P^2, \Rightarrow$ ~~$x \in P^2$~~
or x is point at infinity.

11 $\forall$ line $L \in P^2 \Rightarrow L$ is line in R^2

or L is line at infinity.

here $P^2 = \{ \text{homogeneous vector } v \mid v \in R^3 - \{0, 0, 0\}^T \}$

2.5

$\therefore$ point $\in P^2$,
line $\in P^2$.

3. point $x \in \text{line } L$

$$\Leftrightarrow xL = 0$$

4. dof: degrees of freedom.

$$\text{dof}(\text{point}) = 2$$

$$\text{dof}(\text{line}) = 2.$$

5. $x \in L, x' \in L'$

$$\Leftrightarrow x = L \times L'.$$

here x is cross product.

$$(x_1, x_2, x_3)^T \times (y_1, y_2, y_3)^T$$

$$= \begin{vmatrix} i & j & k \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

$$= \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} i + \begin{vmatrix} -x_1 & x_3 \\ -y_1 & y_3 \end{vmatrix} j + \begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} k$$

Chapter 2. HZ

~~2.57-2.79~~

1. plane $\Leftrightarrow R^2$

prove: point $P \in \text{plane} \Leftrightarrow (x, y) \in R^2$

2. projective space P^2

$:=$ the set of equivalence classes of vectors in $R^3 - (0, 0, 0)^T$

$\Leftrightarrow \{ \text{homogeneous vector } v \mid v \in R^3 - \{0, 0, 0\}^T \}$
2.2 homogeneous vector v is a equivalence class.
2.3. line in R^2

$$\Leftrightarrow ax + by + c = 0$$

$$\Leftrightarrow k(a, b, c)^T, k \neq 0$$

2.4 point $x \in \text{line } L$

$$\Leftrightarrow ax + by + c = 0$$

$$\Leftrightarrow k(x, y, 1) \cdot L = 0, \forall k \neq 0$$

15.3.

$$\forall z \in L \Rightarrow \vec{l}^T z = 0$$

$\therefore \vec{l}^T x = 0, \vec{l}^T y = 0, x \neq y$, i.e., x, y linearly independent, representation of a conic.

$$\therefore \exists \alpha, \text{ s.t. } z = \alpha x + y$$

15.4 $\therefore$ ~~line~~

$$\therefore \forall z \in \text{line } L \Leftrightarrow z = \alpha x + y$$

16. line L tangent to conic C at point x on C

$$\Rightarrow L = Cx.$$

17. dual conics are also known as conic envelopes.

18. point conic.

line conic

degenerate conics.

C is a homogeneous

14.4 $\deg(\text{conic}) = 5$

12.3

such a ray may be thought of as representing a single point x in P^2 .

12.4

In this model, the lines in P^2 are planes passing through the origin.

$$(\text{use } L = x \times x) \text{ (or use 15)}$$

14.5. 15.

$$15.2 \ x \in L, y \in L$$

$$\Rightarrow x + \alpha y \in L, \forall \alpha$$

11. the study of P^2 is

known as projective geometry.

12. P^2 is a set of rays in R^3 .

13. duality principle:

to any theorem of 2d projective geometry there corresponds a dual theorem, which may be derived by interchanging the roles of points and lines in the original theorem.

14 the equation of a conic in

inhomogeneous coordinates is

$$ax^2 + bxy + cy^2 + dx + ey + f = 0$$

i.e., a polynomial of degree 2.

14.3 homogenizing this

$$\Rightarrow x^2 Cx = 0, C = \begin{pmatrix} a & b/2 & c \\ b/2 & e & f \\ c & f & f \end{pmatrix}$$

and the dof of it is 0.

23.3 an alternative definition of a projective transformation:

prove:

$$\text{let } \vec{l}^T x_i = 0, \forall i \in \{1, 2, 3\}$$

A planar projective transformation

$$\therefore \vec{l}^T H^{-1} H x_i = 0$$

is a linear transformation on

$$\therefore (H^{-1} \vec{l})^T (H x_i) = 0$$

homogeneous 3d vectors

$$\therefore H x_i, i=1, 2, 3 \text{ lie on the}$$

represented by a non-singular

$$\text{line } H^{-1} \vec{l}.$$

3x3 matrix.

$$x' = Hx.$$

23.4 H is a homogeneous matrix.

$$\Leftrightarrow \exists H \text{ non-singular}$$

$$\text{such that } \forall x \in \mathbb{P}^2$$

$$Hx = Hx.$$

(E)

24. Projection along rays through a common point defines a mapping from one plane to another.

The central projection mapping

may be expressed by $x' = Hx$ where

H is invertible.

If we define Euclidean coordinate

systems in the two planes,

then it is called a perspective

rather than a full projectivity.

$$= 8$$

19. a projectivity is an invertible mapping h from $\mathbb{P}^2$ to itself such that points x_1, x_2, x_3 lie on the same line $\Leftrightarrow$ if $h(x_1), h(x_2), h(x_3)$ do.

19.3 A projectivity is also called a collineation, a projective transformation, or a homography.

19.4 projectivities form a group

19.5 $\mathbb{P}^2$ is homogeneous 3d vectors

20. x_1, x_2, x_3 lie on same line

H non-singular

$$\text{dof (projective transformation)} \Rightarrow Hx_1, Hx_2, Hx_3 \text{ lie on same line.}$$

31.5 the Euclidean group is a subgroup of the affine group for which the upper left 2×2 matrix is orthogonal.

Oriented Euclidean group is a subgroup of the Euclidean group in which the upper left 2×2 matrix has determinant 1.

31.6 perspectivities are not group. (not closed).

32. isometries are transformations of $\mathbb{R}^2$ that preserve Euclidean distance.

~~It is also called Euclidean transformation~~

32.3 $x' = Hx = \begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix} x$,

where R is 2×2 rotation matrix. (i.e., R is an orthogonal matrix).

32.4 It's also known as a displacement.

29. the concatenation of two projective transformations ~~is~~ is a projective transformation. 25.3 (C-) (group)

30. under a point transformation $x' = Hx$, a conic C transforms to $C' = H^{-T}CH^{-1}$. C

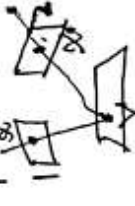
31. $GL(n)$: the group of invertible $n \times n$ matrices with real elements 26. is the real general linear group on n dimensions.

31.3 $PL(n)$: projective linear group, is a quotient group of $GL(n)$. 27.

31.4. affine group is the subgroup of $PL(3)$ consisting of matrices 28. for which the last row is $(0, 0, 1)$.

25. removing the projective distortion from a perspective image of a plane. Note that since the ground and the front are not in the same plane, the projective transformation that must be applied to rectify the front is not the same as the one used for the ground.

projective transformations are important mappings representing many more situations than the perspective imaging of a world plane.



covariant, contravariant behaviour.

point transformation $x' = Hx$

$\Rightarrow$ line transformation $l' = l^T H^{-1}$. C

35. an affine trans is a non-singular linear trans followed by a translation.

$$35.2 \quad x' = T_A x = \begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix} x$$

$$35.3 \quad \therefore A = UDV^T \quad (\text{SVD})$$

$$\therefore A = (UV^T)(VDV^T) = R(\theta)R(-\phi)DR(\phi)$$

($\because U, V$ orthogonal)
 $(D = \begin{pmatrix} \lambda_1 & \lambda_2 \end{pmatrix})$

$\therefore A$ is seen to be the concatenation of a rotation by ϕ , a scaling by λ_1, λ_2 in the rotated x and y directions, a rotation back, and finally another rotation.

35.4 The essence of an affinity is this scaling in orthogonal directions oriented at a particular angle.

33 similarity transformation.

32.5 An isometry:

It's an isometry composed with $(x') = \begin{pmatrix} \epsilon \cos \theta & -\sin \theta & t_x \\ \epsilon \sin \theta & \cos \theta & t_y \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$
 an ~~isometry~~ isotropic scaling.

$$\epsilon = \pm 1$$

33.3.

In the case of a Euclidean transformation composed with a scaling the similarity Euclidean transformation.

has the form:

$$x' = \begin{pmatrix} sR & t \\ 0^T & 1 \end{pmatrix} x$$

32.6 error in 32;

33.4 A similarity trans is also isometries is also called known as an equi-form trans, (because only true when $\epsilon=1$) because it preserves shape. 32.7

34. metric structure implies

that the structure is 32.8 ~~error~~ in 32.4.

defined up to a similarity. It means Euclidean trans.

$\therefore$ the scaling cancels out ~~for~~ for a ratio of areas.

35.6 an affinity is orientation-preserving or -reversing according to whether

$|A| > 0$ or $|A| < 0$ respectively.

36. Projective transformations.

36.2 $x' = H_P x = \begin{pmatrix} A & t \\ v^T & v \end{pmatrix} x$

where $v = (v_1, v_2)^T$

37. $H_A \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} A & t \\ v^T & v \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ 0 \end{pmatrix}$

$H_P \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} A & t \\ v^T & v \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ v x_1 + v x_2 \end{pmatrix}$

$\therefore$ ideal point is mapped to a finite point under H_P .

$\therefore H_P$ can model vanishing points.

here α is the angle between the line and the x -axis of the orthogonal scaling direction.

$\therefore$ we get scaling magnitude λ .
 $\therefore \lambda$ is common to all lines with the same direction.

$\therefore$ it cancels out in a ratio of parallel segment lengths.

(iii) $\therefore$ from 35.3,

rotations and translations do not affect area, so only the scalings by λ_1 and λ_2 matter.

$\therefore$ area is scaled by $\lambda_1 \lambda_2$ which is equal to $|A|$.

$\therefore$ the area of any shape is scaled by $|A|$.

6

35.5

Under an affinity,

three important invariants are

(i) parallel lines.

(ii) ratio of lengths of parallel line segments.

(iii) ratio of areas.

prove:

(i) L_1, L_2 parallel,

$\therefore$ they intersect at $(x_1, x_2, 0)$

$\therefore H_A \begin{pmatrix} x_1 \\ x_2 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1' \\ x_2' \\ 0 \end{pmatrix}$

$\therefore L'_1, L'_2$ intersect at infinity

$\therefore L'_1, L'_2$ parallel.

(ii) $\frac{1}{\cos \alpha} \cos \alpha \quad \lambda^2 = (\lambda_1 \cos \alpha)^2 + (\lambda_2 \sin \alpha)^2$

$\therefore \lambda = \sqrt{(\lambda_1 \cos \alpha)^2 + (\lambda_2 \sin \alpha)^2}$

4). topology of the projective plane

41.3 the projective plane may be pictured as the unit sphere with opposite points identified.

i.e. a line in P^2 is modelled as a great circle on the unit sphere with opposite points identified.

41.4. the sphere S^2 is a 2-sheeted covering space of P^2 .

i.e. P^2 is not simply-connected.

41.5 the fundamental group of P^2 is the cyclic group of order 2.

41.6 P^2 is simply a disk with opposite points on its boundary identified, or glued together.

41.7 P^2 is non-orientable.

41.8 P^1 is topologically equivalent to a line segment with the two endpoints identified — namely a circle, S^1 .

40 the cross ratio is the basic projective invariant of P^1 .

40.2 given 4 points x_i the cross ratio is defined as

$$\text{Cross}(x_1, x_2, x_3, x_4)$$

$$= \frac{|x_1 x_2| |x_3 x_4|}{|x_1 x_3| |x_2 x_4|}$$

$$\text{where } |x_i x_j| = \det \begin{pmatrix} x_{i1} & x_{j1} \\ x_{i2} & x_{j2} \end{pmatrix}$$

40.3 Under a projective transform of the plane, a 1D projective transform is induced on any line in the plane.



40.4

38.

A projective transformation H

$$\Rightarrow H = H_S H_A H_P$$

$$= \begin{pmatrix} sR & t \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} K & 0 \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ v^T & v \end{pmatrix}$$

$$= \begin{pmatrix} A & t \\ v^T & v \end{pmatrix}$$

with A a nonsingular matrix given by $A = sRK + tv^T$, and K an upper-triangular matrix normalized as $|K| = 1$.

This decomposition is valid provided $v \neq 0$, and is unique if s is chosen positive.

$$39, H_E < H_S < H_A < H_P$$

$$\deg(H_E) = 3, \deg(H_S) = 4, \deg(H_A) = 6, \deg(H_P) = 8$$

to determine points and line at infinity.

46. three collinear points a, b, c in an image corresponding to collinear world points with internal ratio $a:b$.

$\Rightarrow$ vanishing point on the line abc

47. homogeneous coordinate $(x, y, 0)$

$\Leftrightarrow$ limit point, $\lim_{\lambda \rightarrow 0} (x, y, \lambda)$

48. The circular points I, J , also called the absolute points, are a pair of complex conjugate ideal points.

The canonical coordinates are:

$$I = \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}, J = \begin{pmatrix} 1 \\ 0 \\ -i \end{pmatrix}$$

49. The circular points I, J are fixed points under the projective trans $H \Leftrightarrow H$ is a similarity.

prove: $\textcircled{1} \Leftarrow \textcircled{2} \Rightarrow$ later.

(A reflection swaps I and J . $\textcircled{2}$)

prove: $\textcircled{1}$

method 1:

$$x = \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}$$

let $x_1 \neq x_2, x_1, x_2 \in L$.

$\Rightarrow Hx_1, Hx_2 \in L$, and $Hx_1 \neq Hx_2$. Affine rectification.

$\therefore$ line L is mapped to L_∞ .

method 2:

$$H^{-T}L = H_A^{-T} \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ i & -i \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \\ i \end{pmatrix}$$

$$= H_A^{-T} \begin{pmatrix} 1 & -1 \\ 0 & 1 \\ 1/i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ i \end{pmatrix}$$

$$= H_A^{-T} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = L_\infty$$

45. Affine properties may be recovered by simply specifying a line (2 dof). Conversely if affine properties are known, these may be used

8

42. the line at infinity, L_∞ is a fixed line under the projective trans $H \Leftrightarrow H$ is an affinity. $\textcircled{1}$

$\textcircled{2}$ H_1, H_2 projective transformations.

$\textcircled{3}$ H_1 maps L_∞ from $(0, 0, 1)^T$ on Euclidean plane π_1 to a finite line L on the plane π_2 .

$\textcircled{4}$ H_2 maps L back to $(0, 0, 1)^T$ on plane π_3 .

$\Rightarrow H_1 H_2$ is affine trans from π_1 to π_3 .

$\textcircled{5}$

44. line $L = (L_1, L_2, L_3)^T, L_3 \neq 0$

H_A is any affine trans.

$$H = H_A \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ L_1 & L_2 \end{pmatrix}$$

$\Rightarrow H$ maps L to $L_\infty = (0, 0, 1)^T$

54. a conic is determined uniquely (up to scale) by five points in general position. Similarly, five lines in general position define a dual conic.

55. The dual conic C^* is fixed under the projective trans $H \Leftrightarrow H$ is a similarity. (see 49)

$$55.3 \quad C^*_{\infty} = IJ^T + JI^T$$

C^*_{∞} is a ~~degenerate~~ degenerate line conic. (rank 2), which consists of the two circular points.

55.4 $C = l m^T + m l^T$ is composed of two lines l, m .
56 degenerate point conics include two lines (rank 2), and a repeated line (rank 1).

⑤ degenerate line conics include two points (rank 2), and a repeated point (rank 1).

53.4 dual (line) conic is a conic which defines an equation on lines.

53.5

point conic C
dual conic C^*

C^* is cofactor matrix of C ,
i.e. C symmetric,

51.3 if C^* is adjoint matrix of C .

53.6

for non-singular symmetric

matrix $C \Rightarrow C^* = C^{-1}$ (up to scale) $\Rightarrow \text{adj}(M) \cdot M = M \cdot \text{adj}(M) = |M| I$

53.7

the line l tangent to conic

satisfies $l^T C^* l = 0$

50. cross product

$$a \times b = [a]_x b = (a^T [b]_x)^T$$

where $[a]_x := \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$ is skew-symmetric matrix.

51. matrix M is a square matrix.

M^* is cofactor matrix of M .

$\text{adj}(M)$ is (classical) adjoint of M .

51.3 if M invertible,

$$\Rightarrow M^* = |M| M^{-T}$$

52. $[a]_x$ is singular, a is its null-vector (right or left).

53.3 point conic is a conic

which defines an equation on points

$$\cos \theta = \frac{l^T C_{\infty}^* m}{\sqrt{l^T C_{\infty}^* l} \sqrt{m^T C_{\infty}^* m}} \quad (1)$$

prove: let $l = (l_1, l_2, l_3)^T$, $m = (m_1, m_2, m_3)^T$

$$\textcircled{1} \quad \cos \theta = \frac{l_1 m_1 + l_2 m_2}{\sqrt{l_1^2 + l_2^2} \sqrt{m_1^2 + m_2^2}} \quad (2)$$

$$\therefore C_{\infty}^* = IJ^T + JI^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\therefore$ (1) reduces to (2) in a Euclidean coordinate system.

$$\textcircled{2} \quad l' = H^{-T} l, \quad m' = H^{-T} m, \quad C_{\infty}^* = H C_{\infty}^* H^T$$

$$\therefore l^T C_{\infty}^* m' = l^T H^{-T} \cdot H C_{\infty}^* H^T \cdot H^{-T} m = l^T C_{\infty}^* m$$

$\textcircled{3}$ similarly for $l^T C_{\infty}^* l'$, $m^T C_{\infty}^* m'$.

$\textcircled{4}$ the scale of the homogeneous objects cancels between the numerator and denominator in (1).

$\therefore$ (1) is invariant to projective trans.
 $\therefore$ proved.

prove: $\therefore x = H^{-1} x'$

$$x^T C_{\infty} x = x'^T H^{-T} C_{\infty} H^{-1} x'$$

$$\therefore C' = H^{-T} C_{\infty} H^{-1}$$

61. Under a point trans $x' = Hx$, a dual conic C^* transforms to

$$C^{*'} = H C^* H^T$$

proof:

$$\therefore l' = H^{-T} l$$

$$\therefore l = H^T l'$$

$$\therefore l^T C^* l = l'^T H C^* H^T l'$$

$$= l'^T H C^* H^T l'$$

$$\therefore C^{*'} = H C^* H^T$$

62. Once the conic C_{∞}^* is identified on the projective plane then Euclidean angles may be measured by

57. circular points arise because every circle intersects ∞ at the circular points. $\therefore$

58. Algebraically, the circular points are the orthogonal directions of Euclidean geometry, $(1, 0, 0)^T$ and $(0, 1, 0)^T$, packaged into a single complex conjugate entity,

$$\text{e.g. } I = (1, 0, 0)^T + i(0, 1, 0)^T$$

$\therefore$ It's not so surprising that once the circular points are identified, orthogonality, and other metric properties, are then determined.

59 C_{∞}^* has 4 degrees of freedom. $\therefore$

$\textcircled{2}$ ∞ is the null vector of C_{∞}^* . $\therefore$

60 under a point trans $x' = Hx$,

a point conic C transforms to $C' = H^T C H$ (see 30)

Singular values are real and non-negative. that maps the imaged circular points 63. (see 55) C_{∞}^* fixed under projective trans H to their canonical position on l_{∞} . $\Rightarrow H$ is a similarity. $\Rightarrow H$ is a similarity.

proof: $A^T A$ symmetric and $\#$ positive-semi-definite.

$\therefore$ its eigenvalues are real and non-negative

$\therefore$ proved. (here see 67)

69. A real symmetric,

$\Rightarrow A = UDU^T$, where U orthogonal, 66. A square, $\Rightarrow A = UDV^T$,
 D real diagonal.

$\therefore$ a real symmetric matrix has real eigenvalues, and the eigenvectors are orthogonal.

70. Once C_{∞}^* is identified on the projective plane, then projective distortion may be the square-roots of $A^T A$ rectified up to a similarity.

66.3 SVD exists for non-square A .

67. the singular values of A are the square-roots of the eigenvalues of $A^T A$

proof: $A = UDV^T$

$$\therefore A^T A = VDU^T UDV^T = VD^2 V^T$$

$\therefore D^2$ are eigenvalue matrix of $A^T A$.

$\Rightarrow$ the trans between the world ~~and~~ plane and the rectified image is a similarity. $H_S C_{\infty}^* H_S^T$

(1) $\Leftarrow$

and the rectified image is a similarity. $H_S C_{\infty}^* H_S^T$

proof: ① the trans is projective

② the circular points are fixed.

$\therefore$

$$= \begin{pmatrix} sR & t \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} sR^T & 0 \\ t^T & 1 \end{pmatrix}$$

$$= \begin{pmatrix} sR & t \\ 0^T & 1 \end{pmatrix} \begin{pmatrix} sR^T & 0 \\ 0^T & 1 \end{pmatrix}$$

$$= s^2 \begin{pmatrix} 1 & 0 \\ 0^T & 1 \end{pmatrix} = s^2 C_{\infty}^* = C_{\infty}^* \text{ (up to scale)}$$

U, V orthogonal matrices, D diagonal matrix with non-negative entries. $\therefore C_{\infty}^*$ is fixed.

(SVD)

(2) $\Rightarrow$ later.

64. C_{∞}^* is dual to the circular points.

(E)

65. If the circular points are identified in an image, and the image is then rectified by a projective trans H

76. dual conjugacy relationship for lines:

B3 conic C , point x
 $\Rightarrow$ polar $l = Cx$.

two lines l and m are conjugate if $l^T C^* m = 0$

77. A symmetric,

$\Rightarrow \exists T$ involtile, s.t. $A = T^T D T$

$D = \text{diag}(\epsilon_1, \dots, \epsilon_n)$ 74. A conic is an involtile

here $\epsilon_i \in \{1, -1\}$

$\epsilon_i = 1, \epsilon_j = -1$ if $i < j$, and one is 1, the other is -1.
 $\epsilon_i = -1, \epsilon_j = 0$, if $i < j$, and one is -1, the other is 0.
 $\epsilon_i = 1, \epsilon_j = 0$, if $i < j$, and one is 1, the other is 0.

Values $\epsilon_i = \pm 1$ occur before values $\epsilon_i = -1$ which in turn precede $\epsilon_i = 0$.

proceeds $\epsilon_i = 0$.
 proof: $\because A = U^T D U$, orthogonal D real diagonal.
 $D = T^T D T, \therefore A = T^T D T, T = U U^T$

71. The polar line $l = Cx$ of point x w.r.t conic C intersects the conic in two points. The two lines tangent to C at these points intersect at x .

proof: $y \in C$,
 $x \in$ tangent line Cy
 $\Leftrightarrow x^T C y = 0$
 $\Leftrightarrow (Cx)^T y = 0$
 $\Leftrightarrow y \in \text{line } l$.

72. If point x is on C then the polar is the tangent line to the conic at x .

B3. the conic induces a map between points and lines of P^2 .

The map is a projective construction since it involves only intersections and tangency, both properties that are preserved under projective transformations.

mapping from points of P^2 to lines of P^2 . It's represented by a 3×3 non-singular matrix A as $l = Ax$.

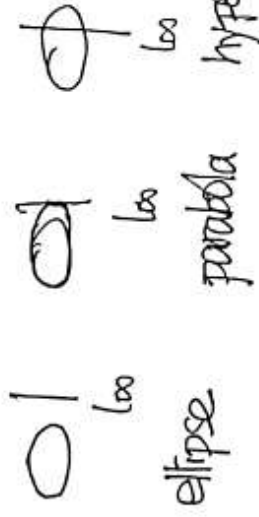
If point x, y satisfying $y^T C x = 0$ then x, y are conjugate w.r.t C .

12. If point x is on C then the polar is the tangent line to the conic at x .

B3

12

$\therefore$ the classification is unaltered by an affinity.



8D.4.

The three types of conic are projectively equivalent to a circle.

8I. fixed points and lines.

8I.3. Eigenvector corresponds to a fixed point of the transformation.

$$He = \lambda e.$$

e and λe represent the same point.

8I.4 $\therefore l' = H^{-T}l$, $\therefore$ fixed lines correspond to the eigenvectors of H^T .

8I.5. line is fixed as a set, $\therefore$ line is fixed pointwise.

②. $x \in C$,

l tangent to C at x

$$\therefore l = Cx$$

$$\therefore C' = H^{-T}CH^{-1}$$

$$x' = Hx, \quad l' = H^{-T}l$$

$$\therefore C'x' = H^{-T}Cx = H^{-T}l = l'$$

$\therefore l'$ tangent to C' at x' .

8D. Non-degenerate, proper conics in Euclidean geometry ~~include~~ are ellipse, parabola, ~~and~~ or hyperbola.

8D.3 $\therefore$ the Euclidean classification depends only on the relation of l_∞ to the conic.

E $\therefore l_\infty$ is fixed under affine trans, $\therefore l'^T x' = [H^{-T}H]x = l^T x = 0$

and intersections are preserved ₁₃

78. Any conic is equivalent under projective transformation to one with a diagonal matrix D . $\therefore$

$\therefore$ The types of conics may now be enumerated:

$$D = \text{diag}(1, 1, 1)$$

$$D = \text{diag}(1, 1, -1), \quad \text{circle.}$$

$$D = \text{diag}(1, 1, 0)$$

$$D = \text{diag}(1, -1, 0), \quad \text{two lines } x = \pm y$$

$$D = \text{diag}(1, 0, 0), \quad \text{single line } x = 0 \text{ counted twice.}$$

79. intersections and tangency are preserved under projective transformations.

proof: $\textcircled{1} x \in l \cap m, \therefore x = l'x = m'x =$

$$x' = Hx, \quad l' = H^{-T}l, \quad m' = H^{-T}m$$

$$\therefore x' \in l' \cap m'$$

$$\therefore |M|=0$$

$$\therefore x \cdot (x_1 \times x_2) = 0$$

$$\therefore \text{line } L = x_1 \times x_2.$$

7. In P^3 , plane π passes x_1, x_2, x_3 ,

$$\Rightarrow \pi = \begin{pmatrix} (x_1 - x_3) \times (x_2 - x_3) \\ -x_3^T (x_1 \times x_2) \end{pmatrix}$$

$$\text{here } x_i^T = w_i(\tilde{x}_i^T, 1)$$

proof: $\forall x' \in \pi$,

$$\therefore |M|=0, \therefore |M| = |x, x_1, x_2, x_3| = 0$$

$$\therefore \begin{vmatrix} x & x_1 & x_2 & x_3 \\ y & y_1 & y_2 & y_3 \\ z & z_1 & z_2 & z_3 \\ w & w_1 & w_2 & w_3 \end{vmatrix} = 0$$

$$\therefore \begin{vmatrix} x & y & z & w \\ x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \\ x_3 & y_3 & z_3 & w_3 \end{vmatrix} = 0$$

$$\therefore \pi = 0, \pi = (D_{234}, -D_{134}, D_{124}, -D_{123})^T$$

$$\text{here } P_{34} = \begin{vmatrix} x & y & z & w \\ x_1 & y_1 & z_1 & w_1 \\ x_2 & y_2 & z_2 & w_2 \\ x_3 & y_3 & z_3 & w_3 \end{vmatrix} = ((y_1 - y_3)(z_2 - z_3) - (y_2 - y_3)(z_1 - z_3)),$$

similarly, other components computed.

Chapter 3

H2

3.3. homogenizing it by replacing

$$X \Rightarrow X_1/X_4, Y = X_2/X_4, Z = X_3/X_4$$

$$\Rightarrow \pi^T X = 0.$$

$$\text{where } \pi = (\pi_1, \pi_2, \pi_3, \pi_4)^T.$$

$$X = (X_1, X_2, X_3, X_4)$$

3.4 the distance of the plane $X' = HX$.

from the origin is $\frac{|d|}{\|n\|}$.

$$4. |A| = |A^T|$$

$$5. a \cdot (b \times c)$$

$$= \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

6. In P^3 , L passes $x_1, x_2, \Rightarrow L = x_1 \times x_2$

$$\text{proof: } \forall x \in L, M = (x_1, x_2, x_3)$$

$$= \begin{pmatrix} x & y & z \\ x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \end{pmatrix}$$

Projective geometry and transformations of 3D.

1. A projective transformation acting on P^3

is a linear trans on homogeneous 4d vectors represented by a non-singular 4×4 matrix.

1.3 H is homogeneous.

$$\det(H) = 15.$$

1.4 The map is collineation which preserves incidence relations

2. points and planes are dual in P^3 .

lines are self-dual in P^3 .

3. a plane in 3d space is:

$$n \cdot \tilde{X} + d = 0, \text{ where } n = (\pi_1, \pi_2, \pi_3)^T, \tilde{X} = (X, Y, Z)^T, d = \pi_4$$

plane intersection.

$\because \forall \pi \in \langle P, Q \rangle,$

$$A^T \pi = B^T \pi = 0$$

$\therefore \pi$ containing $\langle W^T \rangle$.

$\therefore$ proved. $\square$

2. the dual representation of a line:

The line is represented as the span of the row space of the 2×4 matrix W^*

$$W^* = \begin{pmatrix} P^T \\ Q^T \end{pmatrix},$$

P, Q plane.

$$\Rightarrow (i) \langle W^*, \pi \rangle = \langle P, Q \rangle$$

= the pencil of planes $\lambda P + \mu Q$ with the line as axis.

(ii) the 2d null-space of W^* is the pencil of points on the line.

$$13. W^* W^T = W \cdot W^* = 0_{2 \times 2} \quad \square$$

10.4. Lines are very awkward to represent in 3d space.

A number of line representations

have been proposed.

11. The line joining A, B (non-coincident)

is represented by the span of the row space of the 2×4 matrix W .

$$W = \begin{pmatrix} A^T \\ B^T \end{pmatrix}$$

$\Rightarrow$ (i) the span of $W^T = (A, B)$ is

the pencil of points $\lambda A + \mu B$ on the line.

(ii) the 2d right null-space of W is the pencil of planes with the line as axis.

proof: (ii) suppose that $\langle P, Q \rangle = \text{null-space}$

$$\therefore WP = 0 \quad \therefore A^T P = B^T P = 0$$

$$\therefore P \text{ is a plane containing } \langle W^T \rangle$$

Similarly, Q containing $\langle W^T \rangle$.

$\therefore$ the line is $\langle W^T \rangle$. is the

8. under the point transform $X' = HX$, a plane transforms as $\pi' = H^T \pi$.

9. $\forall X \in \pi$.

$$\Rightarrow X = Mx$$

where $\pi^T M = 0$, i.e., the columns of the

4×3 matrix M generate the rank 3

null-space of π^T .

9.3 M is not unique, of course.

9.4 This representation is simply the analogue of a line l in P^2 defined as a linear combination of its 2D

null-space as $x = \mu a + \lambda b$, where

$$C^T a = C^T b = 0.$$

a line is defined by the join of two points or the intersection of two planes

$$10.3 \text{ dof}(\text{line}) = 4, \text{ dof}(\text{point}) = 3,$$

$$\text{dof}(\text{plane}) = 3.$$

$$18.3 \text{ rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$$

18.4. $C=AB$, B square, invertible.

$$\Rightarrow \text{rank}(C) = \text{rank}(A).$$

18.5 U, V vector space $\Rightarrow U+V$ is vs.

18.6 $\dim(U+V) \leq \dim(U) + \dim(V)$, here U, V vs.

proof: $\forall x \in U+V, \Rightarrow x = x_u + x_v, x_u \in U, x_v \in V$

$$\therefore x_u + x_v \in \langle U+V \rangle \quad \therefore x \in \langle U+V \rangle$$

$$\therefore U+V \subset \langle U+V \rangle$$

$$\therefore \dim(U+V) \leq \dim \langle U+V \rangle$$

$$= \dim \langle \text{basis } U \cup \text{basis } V \rangle$$

$$\leq \dim U + \dim V$$

18.7 matrix $A, B \Rightarrow \text{rank}(A+B) \leq r(A) + r(B)$

$$\text{proof: } \therefore \langle A+B \rangle \subset \langle A \rangle + \langle B \rangle$$

$$\therefore \dim \langle A+B \rangle \leq \dim(\langle A \rangle + \langle B \rangle)$$

$$\leq \dim \langle A \rangle + \dim \langle B \rangle$$

$\therefore$ proved.

16.11 $\pi = \text{proj of point } X \text{ and line } W$.

$$\Rightarrow \{\pi\} = \text{null space of } M.$$

$$M = \begin{pmatrix} W \\ X^T \end{pmatrix}$$

(2) point $X = \text{intersection of line } W$
with plane π

$$\Rightarrow \{X\} = \text{null space of } M.$$

$$M = \begin{pmatrix} W^* \\ \pi^T \end{pmatrix}$$

17. Plicker representation

joining pairs A, B

a line is represented by a 4×4

skew-symmetric homogeneous matrix L .

$$L = AB^T - BA^T.$$

~~here the line joining points A, B is represented by L .~~

14. example:

X-axis:

$$W = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, W^* = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

where $A = \text{origin}$,

$B = \text{ideal point in the X-direction}$

$P = XY\text{-plane}$

$Q = XZ\text{-plane}$.

$$15.3 \dim(\text{range}(A)) = \text{rank}(A)$$

where $A \in \mathbb{R}^{m \times n}$

15.4 row space $\perp$ null space

15.5 rank-nullity theorem

$$A \in \mathbb{R}^{m \times n} \Rightarrow \text{rank}(A) + \text{nullity}(A) = n$$

proof: $\therefore \dim(\text{row space}) = \text{rank}(A)$

$$\dim(\text{row space}) + \dim(\text{null space}) = n \quad (15.4)$$

$\therefore$ proved.

25. Under the point transformation 21 . dual Plucker representation L^* . 19 properties of Plucker representation L .
 $X \mapsto HX$, L^* transforms as $L^* = H^{-T} L^* H^T$. $L^* = PQ^T - QP^T$. $19.3 \text{ rank}(L) = 2$.

26. L^* can be obtained directly here the line is formed by the intersection of two planes P, Q . $\textcircled{2} L W^{*T} = 0 \in \mathbb{R}^{4 \times 2}$

here $w^* = (P, Q)$, P, Q planes.

see 12.(ii), vs 19.3⑤

$$19.5 \text{ dof}(L) = 4$$

19.6 matrix L is independent of the points

A, B used to define it.

19.7. Under point trans $X' = HX$,
 $\Rightarrow L' = HLH^{-1}$.

proof: $L' = A'B^T - B'A^T$

$$= \text{HAB}^T \text{H}^T - \text{HBA}^T \text{H}^T$$

$$= +12H^7$$

20. example:

5. X-axis is represented as: $L = AB^T - BA^T = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$
where A origin, B ideal point in the X-direction.

25. Under the point transformation $\mathcal{Z}$, dual Plücker representation L^*

$X = HX$, L^* transforms as $L^* = H^{-T} L^* H^{-T}$. $L^* = Q^T - Q^T$

26. L^* can be obtained directly here the line is for of two planes P, Q .

from L by the simple rule: 22. correction:

$$l_{12} : l_{13} : l_{14} : l_{23} : l_{42} : l_{34} = l_{34}^* : l_{42}^* : l_{23}^* : l_{14}^* : l_{13}^* : l_{12}^* \quad (X)$$

$$\pi' = H^{-T} \pi \quad (v)$$

27. A skew-symmetric matrix

$$\Rightarrow \text{rank}(A) \neq 1.$$

proof: $\exists u, v \neq 0, \text{ s.t. } A u v^T H^{-T} (x_1 \times x_2) = f(x_1) \times f(x_2)$

proof:

$$\therefore l' = H^{\perp} L, \quad l' = x' x x' x'$$

$$l = x_1 \times x_2$$

$$\therefore (Hx_1) \times (Hx_2) = H^{-T}(x_1 \times x_2)$$

24. M is any 3×3 matrix, $\mathcal{L}, \mathcal{V}$ vectors. example:

$$\Rightarrow M^*(x, y) = (M_x) \times (M_y)$$

$i, kv^T = 0 \Rightarrow v^T = 0$, error. is proved. where M^* is cofactor matrix of M . $\square$

proof:

$$\textcircled{1} \text{ let } \pi = L^*X, L^*X \neq 0$$

$$\because L^* = PQ^T - QP^T \quad \therefore L \in \pi$$

$$\textcircled{2} \because X^T \pi = X^T L^*X$$

$$= X^T PQ^T X - X^T QP^T X$$

$$= (X^T Q)(X^T P) - (X^T Q)(X^T P)$$

$$= 0$$

$$\therefore X \in \pi$$

$$\textcircled{3} \because L^*X \text{ is the plane } \pi \text{ when } L^*X \neq 0$$

$$\textcircled{4} L^*X = 0 \Leftrightarrow P^T Q^T X - Q^T P^T X = 0$$

$$\Leftrightarrow Q^T X = P^T X = 0$$

$$\Leftrightarrow X \in Q, X \in P$$

$$\Leftrightarrow X \in L$$

proof:

$$\text{let } C = A + \mu B$$

$$\therefore L = AC^T - CA^T$$

$$= AA^T + \mu AB^T$$

$$- AA^T - \mu BA^T$$

$$= \mu (AB^T - BA^T)$$

$$= \mu L$$

$$\therefore \hat{L} = L \text{ (up to scale)}$$

proof 2:

(later)

30. plane π is defined by the join of point X and line L .

$$\Rightarrow \pi = L^*X, \text{ when } L^*X \neq 0, 29.$$

$$\text{and } L^*X = 0 \Leftrightarrow X \in L$$

28.3

$$L = AB^T - BA^T, A, B \text{ different points.}$$

$$\Rightarrow L \neq 0$$

proof: If $L = 0$

$$\Rightarrow AB^T = BA^T$$

$$\Rightarrow A, B \text{ linear dependent}$$

$$\Rightarrow A, B \text{ same point.}$$

$$\therefore L \neq 0$$

$$28.4 \quad \text{rank}(L) = 2$$

proof:

$$L \text{ skew-symmetric, } L \neq 0$$

$$\Rightarrow r(L) \geq 2$$

$$\Rightarrow r(L) = 2 \quad (\because L = AB^T - BA^T)$$

$$\therefore r(L) \leq 2$$

L is independent of A, B . (19.6)

36. proof 2 of 29. (see 19.6)

$$\therefore M' = (A', B'), M = (A, B)$$

$$\therefore M' = M\Lambda$$

$$\therefore L' = L|\Lambda|$$

$$\therefore L' = L \text{ (up to a scale)}$$

$$37. \therefore \text{Rank}(M) = 2$$

$$\therefore L \neq 0$$

$$34. \text{ let } M = (A, B)$$

$\therefore L_{ij}$ is the determinant of rows i and j of M .

proof:

$$\therefore L = AB^T - BA^T$$

$$\therefore L_{ij} = b_j a_i - a_j b_i$$

$$= \begin{vmatrix} a_i & b_i \\ a_j & b_j \end{vmatrix}$$

$$\text{here } A = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix}, B = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \end{pmatrix}$$

$$35. \therefore L_{ij} = -L_{ji}$$

$\therefore$ Plücker line coordinates have $C_4^2 = 6$ elements.

31. point X is defined by the intersection of line L with plane π .

$$\Rightarrow X = L\pi, \text{ when } L\pi \neq 0$$

$$\text{and } L\pi = 0 \Leftrightarrow L \in \pi$$

32. e.x.,

the intersection of the X -axis with plane $X=1$ is given by $X=L\pi$ as

$$X = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

33. Plücker line coordinates are the

six non-zero elements of Plücker matrix

$$L: L = \{l_{12}, l_{13}, l_{14}, l_{23}, l_{24}, l_{34}\}$$

33.3

$$|L| = 0 \Rightarrow l_{12}l_{34} + l_{13}l_{42} + l_{14}l_{23} = 0$$

(3)

41. line L : join of A, B .
 line $\hat{L}$: join of $\hat{A}, \hat{B}$.

L and $\hat{L}$ intersect (i.e., coplanar)

$\Leftrightarrow A, B, \hat{A}, \hat{B}$ coplanar.

$\Leftrightarrow |A, B, \hat{A}, \hat{B}| = 0$

$\Leftrightarrow (L|\hat{L}) = 0$

proof:

$\therefore$ the bilinear product

$$(L|\hat{L}) := l_{12}\hat{l}_{34} + l_{13}\hat{l}_{42} + l_{14}\hat{l}_{23} + l_{23}\hat{l}_{14} + l_{42}\hat{l}_{13} + l_{34}\hat{l}_{12}$$

$$\therefore (L|\hat{L}) = |A, B, \hat{A}, \hat{B}| \quad (\text{from } 40)$$

$\therefore$ proved.

$$= (a-b+c)^2$$

$$\text{here } a = l_{12}l_{34}, b = l_{13}l_{42}, c = l_{14}l_{23}$$

$$\therefore |L| = 0 \Leftrightarrow a - b + c = 0$$

$$\Leftrightarrow l_{12}l_{34} + l_{13}l_{42} + l_{14}l_{23} = 0$$

$$40. \begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix}$$

$$= k_{12}l_{34} + k_{13}l_{42} + k_{14}l_{23} + k_{23}l_{14} + k_{24}l_{13} + k_{34}l_{12}$$

$$\text{where } k_{ij} = \begin{vmatrix} a_i & b_i \\ a_j & b_j \end{vmatrix}$$

$$l_{ij} = \begin{vmatrix} c_i & d_i \\ c_j & d_j \end{vmatrix}$$

38. Leibniz formula:

$A: n \times n$

$$\Rightarrow |A| = \sum_{\sigma \in S_n} \text{sign}(\sigma) \prod_{i=1}^n A_{i\sigma(i)}$$

where S_n is the set of all permutations of the integers $\{1, 2, \dots, n\}$.
 $\sigma(i)$ is the element in position i after the reordering σ .

$$\text{sign}(\sigma) = \begin{cases} 1, & \sigma \text{ is even} \\ -1, & \sigma \text{ is odd} \end{cases}$$

38.3 σ is even (odd) if the new sequence can be obtained by an even (odd) number of switches of numbers.

39. prove 33.3: $\therefore$

$$\text{proof: } |L| = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$$

③. in ②, we supposed $q_{12} \neq 0$,

If $q_{ij} \neq 0, i < j$, then let $m, n \in \{1, 2, 3, 4\} / \{i, j\}$
with $m < n$.

then M is defined by:

$$\text{row } i: \quad q_j \quad 0$$

$$\text{row } j: \quad 0 \quad q_i$$

$$\text{row } m: \quad -q_m \quad q_{im}$$

$$\text{row } n: \quad -q_n \quad q_{in}$$

$\therefore l_{st} = q_{ij} q_{st}$, when $(s, t) \neq (m, n)$

$$l_{mn} = -q_{jm} q_{in} + q_{jn} q_{im} = q_{ij} q_{mn}$$

($\because q_{ij} q_{mn} + q_{jm} q_{in} - q_{jn} q_{im} = 0$ by $(Q|Q) = 0$)

$\therefore Q$ is line L .

44. $\therefore$ the lines in P^3 are in bijective correspondence

with Plicker quadric or Klein quadric in P^5 .

(from 43)

proved.

then define $M = (A, B)$

$$= \begin{pmatrix} q_{12} & 0 \\ 0 & q_{12} \\ -q_{23} & q_{13} \end{pmatrix} \quad q_{12} \quad q_{14}$$

$\therefore A, B \in P^3, A \neq B$

$\therefore$ line $L = (l_{12}, l_{13}, l_{14}, l_{23}, l_{24}, l_{34})$

with

$$l_{12} = q_{12}^2, \quad l_{13} = q_{12} q_{13}, \quad l_{14} = q_{12} q_{14}$$

$$l_{23} = q_{12} q_{23}, \quad l_{42} = q_{12} q_{42}, \quad \text{etc.}$$

$$l_{34} = -q_{14} q_{23} - q_{13} q_{42} = q_{12} q_{34} \quad (\because (Q|Q) = 0)$$

$\therefore L = (q_{12}, q_{13}, q_{14}, q_{23}, q_{12}, q_{34})$

up to scale.

$\therefore Q$ is line L .

42. $L = AB^T - BA^T$

$$\Rightarrow l_{12} l_{34} + l_{13} l_{42} + l_{14} l_{23} = 0$$

(see 33)

proof:

$$L = AB^T - BA^T$$

$$\Rightarrow l_{ij} = \begin{vmatrix} a_i & b_i \\ a_j & b_j \end{vmatrix}$$

$$\Rightarrow (L|L) = |A, B, A, B| \quad (\text{from 40})$$

$$\Rightarrow (L|L) = 0$$

$$\Rightarrow l_{12} l_{34} + l_{13} l_{42} + l_{14} l_{23} = 0$$

43. $L \in P^5$ is a line in P^5

$$\Leftrightarrow (L|L) = 0$$

proof: $\Rightarrow$ (from 42)

$\Leftarrow$: let $Q = (q_{12}, q_{13}, q_{14}, q_{23}, q_{42}, q_{34})^T \in P^5$
($Q|Q$) = 0, suppose $q_{12} \neq 0$

$$Q' = H^{-1} Q H^{-1} \quad (30 \text{ in } C2)$$

446 dual quadrics are equations on planes: the tangent planes π to the point quadric Q satisfy

$\pi^T Q^* \pi = 0$, where $Q^* = \text{adjoint } Q$, or Q^{-1} if Q is invertible.

446.3 Under the point trans $X' = HX$, 458.3 $X \in \pi, X \in Q$

a quadric transforms as $Q' = H Q^* H^T$ $\Leftrightarrow X = Mx, X^T Q X = 0$

47. classification of quadrics (see 78 in C2)

47.3 the projective type of a quadric is uniquely determined by its rank and signature.

47.4 the signature of a diagonal matrix D , $\text{dis}(1, \dots, 1, 0)$, denoted $\sigma(D)$, is defined to be the number of

$+1$ entries minus the number of -1 entries.

47.5 Q real symmetric, $\sigma(Q) := \sigma(D)$, s.t. $Q = H^T D H$, where H is real, D in 47.4 σ is well defined, being independent of H .

45.8 the intersection of a plane π with a quadric Q is a conic C .

A coordinate system for the plane can be defined by

the complement space to π as $X = Mx$. (see 9)

458.3 $X \in \pi, X \in Q$

$\Leftrightarrow X = Mx, X^T Q X = 0$

$\Leftrightarrow X = Mx, x^T C x = 0$,

$$C = M^T Q M$$

$\Leftrightarrow X = Mx,$

$x \in C, C = M^T Q M$.

45.9 Under the point trans $X' = HX$,

$$X' = HX,$$

a (point) quadric transforms as

9

45. A quadric is a surface in P^3 defined by $X^T Q X = 0$, where Q is a symmetric 4×4 matrix.

45.3 Q : matrix Q , or the quadric surface it defines.

45.4 $\det(Q) = 9$, (14.4 in C2)

45.5 nine points in general position define a quadric.

45.6 A quadric defines a polarity

between a point and a plane, in a

similar manner to the polarity defined

by a conic ~~defined~~ between a point

and a line. (73 in C2)

45.7 the plane $\pi = QX$ is the polar

plane of X with respect to Q .

(71 in C2)

(three for rotation, three for translation, one for isotropic scaling), five for affine scalings, and three for the projective part of the trans.

49.3

Group

Matrix

$\text{def}(\text{projective}) = 15$

$$\begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix}$$

$\text{def}(\text{affine}) = 12$

$$\begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix}$$

$\text{def}(\text{similarity}) = 7$

$$\begin{pmatrix} sR & t \\ 0 & 1 \end{pmatrix}$$

$\text{def}(\text{Euclidean}) = 6$

$$\begin{pmatrix} R & t \\ 0 & 1 \end{pmatrix}$$

50. Any particular translation and rotation

is equivalent to a rotation about a

screw axis together with a translation along the screw axis. The screw axis is parallel to the rotation axis.

50.4 a is unit direction of the rotation axis

$$\Leftrightarrow Ra = a$$

48.5 C non-singular cubic twisted

$\Rightarrow C$ is not contained within any plane of P^3 .

It intersects a general plane at three distinct points.

48.6 $\text{def}(C) = 12$.

48.7 there is a unique C through 6 points in general position.

48.8 all non-degenerate twisted cubics are projectively equivalent.

49. The 15 degrees of freedom of a projective trans are accounted for as seven for a similarity

10

47.6 quadrics fall into two classes: ruled and unruled quadrics.

47.7 a ruled quadric is one that contains a straight line.

48. the twisted cubic may be considered to be a 3d analogue of a 2d conic.

48.3 a conic in 2d projective plane may be described as a parametrized curve given by the equation:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = A \begin{pmatrix} 1 \\ \theta_1 \\ \theta_2 \end{pmatrix}, \quad A \in 3 \times 3, \text{ non-singular.}$$

48.4 In an analogous manner, a twisted cubic is defined to be a curve in P^3 given by:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = A \begin{pmatrix} 1 \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{pmatrix}, \quad A \in 4 \times 4, \text{ non-singular.}$$

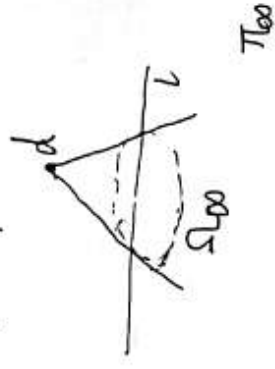
- $\Rightarrow H$ affine
- $\therefore H = H_A = \begin{pmatrix} A^T \\ 0^T \ 1 \end{pmatrix}$
- Restricting Ω_{∞} to the plane at infinity, it is I_{33} .
- $\therefore \Omega_{\infty}$ is fixed by H_A .
- $\therefore A^{-T} I A^{-1} = I$ (up to scale) 54.4 For points with $X_4 = 0$,
- $\therefore A^T A = I$, (up to scale)
- ~~$A^{-1} A$ is orthogonal~~ $\therefore A = sR$, R orthogonal.
- $\therefore H_A$ similarity.
56. Ω_{∞} is only fixed as a set by a general similarity; it is not fixed pointwise.
- 56.4 a line intersects a conic in two points pointwise.
57. The absolute conic Ω_{∞} is a (point) conic on π_{∞} .
- 57.3 In a metric frame $\pi_{\infty} = (0, 0, 0, 1)^T$, the circular points on Ω_{∞} then allows the measurements of metric properties.
- In P^3 line corresponding geometric entities are the plane at infinity π_{∞} and the absolute conic Ω_{∞} .
52. the plane at infinity has the canonical position $\pi_{\infty} = (1, 0, 0, 1)^T$ in affine 3-space.
53. π_{∞} is fixed under the projective transformation $H \Leftrightarrow H$ is an affinity. (42 in C2)
- 53.3 π_{∞} is, in general, fixed as a set, It is not fixed pointwise.
- $\Rightarrow H$ must fix the plane at infinity

57. In P^2 , a conic C and two points, x_1 and x_2 , in general position have the invariant $I = \frac{(x_1^T C x_2)^2}{(x_1^T C x_1)(x_2^T C x_2)}$

Σ (iv) $\text{dual}(P^2)$

58. two directions d_1 and d_2 are orthogonal if $d_1^T \Sigma_{\infty} d_2 = 0$

58.4 A plane normal direction d and the intersection line l of the plane with π_{∞} are in pole-polar relation w.r.t. Σ_{∞}



59. Ω_{∞} is defined by two equations. It's a conic on the plane at infinity.

60. The dual of Ω_{∞} is a degenerate dual quadric in 3d space called the absolute dual quadric, and denoted Ω_{∞}^* .

60.3 Geometrically Ω_{∞}^* consists of the planes tangent to Ω_{∞} , that Ω is the rim of Ω_{∞}^* . This is called a rim quadric.

Algebraically Ω_{∞}^* has the canonical form $\Omega_{\infty}^* = \begin{pmatrix} I_{3 \times 3} & 0 \\ 0^T & 0 \end{pmatrix}$ (up to scale)

61. Ω_{∞}^* is made up of just those planes tangent to the absolute conic.

$\forall \pi = (v^T, k^T) \in \Omega_{\infty}^*$

12

$$\Leftrightarrow \pi^T \Omega_{\infty}^* \pi = 0$$

$$\Leftrightarrow v^T v = 0 \quad (\because \Omega_{\infty}^* = \begin{pmatrix} I & 0 \\ 0^T & 0 \end{pmatrix})$$

$$\Leftrightarrow v^T I v = 0$$

$\Leftrightarrow$ the line represented by v is tangent to the absolute conic

$\Leftrightarrow$ plane π is tangent to the absolute conic.

($\because v$ represents the line in which plane π meets the plane at infinity)

62. absolute conic as the limit of a series of squashed ellipsoids.

$$\Leftrightarrow \text{def}(\Omega_{\infty}^*) = 8$$

63. Ω_{∞}^* fixed under projective trans $H \Leftrightarrow H$ similarity

64. Ω_{∞}^* fixed under projective trans $H \Leftrightarrow H$ similarity

$$\Leftrightarrow \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} = H \begin{pmatrix} I & 0 \\ 0 & 0 \end{pmatrix} H^T = \begin{pmatrix} A^T & A^T v_0 \\ v_0^T & v_0^T v_0 \end{pmatrix} \quad (\text{up to scale})$$

$$\Leftrightarrow v = 0, A \text{ is scaled orthogonal.}$$

$\Leftrightarrow H$ similarity.

65. the plane at infinity π_{∞} is the null-vector of Ω_{∞}^* .

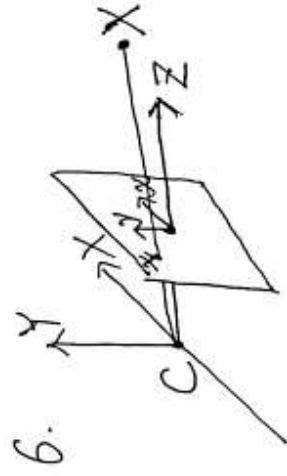
66. the angle between plane π_1 and plane π_2 is given by proof: $\forall \pi = (v^T, k^T) \in \Omega_{\infty}^*$

$$\cos \theta = (\pi_1^T \Omega_{\infty}^* \pi_2) / \sqrt{(\pi_1^T \Omega_{\infty}^* \pi_1)(\pi_2^T \Omega_{\infty}^* \pi_2)}$$

Chapter 6 12 Camera Models

cameras with a finite center, and those that model cameras with center "at infinity".

5.4 affine camera is the natural generalization of parallel projection.



C: camera center, or optical center,
image plane,
principal axis, principal point,
principal plane.

$$(X, Y, Z)^T \rightarrow (fX/Z, fY/Z)^T$$

$$6.4. \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} \rightarrow \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{pmatrix} f & 0 \\ 0 & f \\ 0 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

the camera center in the world coordinate frame,

$$6.5 \quad x = PX,$$

$$P = \text{diag}(f, f, 1) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

X world point, x image point.

7. principal point offset:

$$(X, Y, Z)^T \rightarrow (fX/Z + p_x, fY/Z + p_y)^T$$

where $(p_x, p_y)^T$ are principal point.

$$i, \quad x = K \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x_{\text{cam}}$$

K is called camera calibration matrix.

$$K = \begin{pmatrix} f & p_x \\ 0 & f \\ 0 & 0 & 1 \end{pmatrix}$$

x_{cam} is expressed in camera coordinate frame.

8. camera rotation and translation.

$$\tilde{x}_{\text{cam}} = R(\tilde{X} - \tilde{C}),$$

where $\tilde{C}$ represents the coordinates of the camera center in the world coordinate frame,

1. A camera is a mapping between the 3D world and 2D image.

2. Camera models are matrices with particular properties that represent the camera mapping.

3. The principal camera of interest is central projection.

4. All cameras modelling central projection are specializations of the general projective camera.

The anatomy of this most general camera model is examined using the tools of projective geometry.

5. The specialized models fall into two major classes — those that model

1. α_x, α_y represent the focal length of the camera in terms of pixel dimensions in the x and y direction respectively.

$\tilde{x}_0 = (x_0, y_0)$ is the principal point in terms of pixel dimensions.

9.4 A CCD camera has 10 degrees of freedom.

10. finite projective camera

$$K = \begin{pmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix}$$

s is referred to as the skew parameter.

10.2 A camera $P = KR(I - \tilde{C})$ for which K is of the form above

is called a finite projective camera.

10.3 $\text{doF}(P) = 11$

proof:

$$\tilde{x}_{\text{cam}} = R\tilde{x}$$

$$= A_{W \rightarrow C}\tilde{x}$$

(change of basis)

$$A_{W \rightarrow C} = A_{C \rightarrow W}^{-1}$$

$\therefore A_{C \rightarrow W} = (r_1, r_2, r_3)$

$A_{C \rightarrow W}$ orthogonal

$\therefore A_{W \rightarrow C} = A_{C \rightarrow W}^T = \begin{pmatrix} r_1^T \\ r_2^T \\ r_3^T \end{pmatrix}$

$\therefore R = \begin{pmatrix} r_1^T \\ r_2^T \\ r_3^T \end{pmatrix}$

9. CCD cameras.

$$K = \begin{pmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} m_x \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} f & 0 \\ 0 & f \end{pmatrix}$$

where m_x is the number of pixels per unit distance in image coordinates in x direction. Similarly for m_y

R is 3×3 rotation matrix representing the orientation of the camera coordinate frame. (see 8.6)

$$\begin{aligned} 8.3 \quad x_{\text{cam}} &= \begin{pmatrix} R & -R\tilde{C} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \tilde{x} \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} R & -R\tilde{C} \\ 0 & 1 \end{pmatrix} X \end{aligned}$$

$\therefore x = KR(I - \tilde{C})X$

X is in a world coordinate frame.

8.4 $P = KR(I - \tilde{C})$

$\text{doF}(P) = 9$, ($\because \text{doF}(K) = 3, \text{doF}(R) = 3, \text{doF}(\tilde{C}) = 3$)

8.5 $P = K(R, t)$, where $t = -R\tilde{C}$

8.6 ~~$R = \begin{pmatrix} r_1^T \\ r_2^T \\ r_3^T \end{pmatrix}$~~ $R = \begin{pmatrix} r_1^T \\ r_2^T \\ r_3^T \end{pmatrix}$

r_i is the coordinates of i -th camera coordinate axis. in the world coordinate frame. (here no translation).

1. $\in$ the line $X(A)$

camera center

~~1. A is the camera center~~

1. camera center $\in X_A(\lambda), \forall \lambda$

1. $C \in X_A(\lambda), \forall \lambda$

1. C = camera center.

proof 2:

$$\therefore PC = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is not defined,

1. the camera center is

the unique point in space

for which the image is undefined by the join $X(\lambda) = \lambda A + (1-\lambda)C$.

1. C is camera center.

12.4. finite camera

$$\Rightarrow C = \begin{pmatrix} -M^{-1}P_4 \\ 1 \end{pmatrix}, \text{ where } P = (M, p_4)$$

11. general projective cameras

11.2 A general projective camera is one

represented by an arbitrary homogeneous

3×4 matrix of rank 3.

11.3 It has 11 degrees of freedom.

12. $PC=0$, P is a general projective camera $\Rightarrow C$ is camera center.

proof 1:

(Note C must exist).

$\forall A \in 3d \text{ space } \mathbb{P}^3$.

The line containing C and A is represented

by the join $X(\lambda) = \lambda A + (1-\lambda)C$.

1. $PC=0$

1. all points on the line are mapped

to the same point PA

1. C is camera center.

3

10.4 The set of camera matrices of

finite projective cameras is identical

with the set of homogeneous 3×4

matrices for which the left hand

3×3 submatrix is non-singular.

$$\Rightarrow P = KR(I - \tilde{C})$$

$$= (KR - K\tilde{C}),$$

$\therefore P = (M, p_4)$, M non-singular.

$\Leftarrow$

let $P = (M, p_4)$, M non-singular.

$M = KR$, (RQ matrix decomposition)

($\exists K, R$, s.t. K has form IO, R rotation)

$$P = M(I - M^{-1}p_4) = KR(I - \tilde{C})$$

with $\tilde{C} = -M^{-1}p_4$

15. the image point $x_0 = M m_3$ is the principal point of the camera, where $M = \begin{pmatrix} m_1^T \\ m_2^T \end{pmatrix}$

proof: $\because$ the principal plane is $P_3 = (p_{31}, p_{32}, p_{33})^T$
 $\therefore$ the normal to it is $(p_{31}, p_{32}, p_{33})^T$
 $\therefore$ The normal is represented by a point on the plane at infinity, $(p_{31}, p_{32}, p_{33}, 0)^T = \hat{p}_3$
 $\therefore$ principal point $x_0 = P \hat{p}_3 = (M p_4) \hat{p}_3 = M m_3$.
 16. the principal axis is the ray passing through the camera center C with direction vector m_3^* .
 The principal axis vector $v = \det(M) m_3$ is directed towards the front of the camera.

(see 8.6)

17. $x \rightarrow ray$, s.t. the ray maps to x .

The form of the ray may be specified in several ways, (Plücker representation, join of two points).

17.3. $X(x) = P^T x + \lambda C$, where P^T is pseudo-inverse of P ($P^T = P^T (P P^T)^{-1}$) the image. $\textcircled{3}$ Similarly for P_2 .

17.4 finite camera: $X(u) = u \begin{pmatrix} M^T x \\ 0 \end{pmatrix} + \begin{pmatrix} -M^T p_0 \\ 1 \end{pmatrix}$

$\Rightarrow P_3$ is the principal plane.

12.5 camera at infinity (λ is singular)

P_1, P_2 represent planes in space $\Rightarrow C = (d^T, 0)^T$, where $Md = 0$.

through the camera center, C is a point at infinity.

map to the image lines $x=0$ and $y=0$ respectively.

13. P , a general projective camera

$$P = (p_1, p_2, p_3, p_4)$$

$\Rightarrow p_i$ are the vanishing points of the world coordinate X, Y and

Z axes respectively. ($i=1,2,3$)

$$(\because p_i = P(\hat{p}_i))$$

the principal plane.

p_4 is the image of the world origin.

$$\textcircled{2}. P^T X = 0 \Rightarrow PX = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\because PC = 0 \wedge P^T C = 0 \wedge C \in P_3$$

$\therefore P_3$ is defined by the camera center and the line $x=0$ in

14. P , a general projective camera,

$$P = \begin{pmatrix} P_1^T \\ P_2^T \end{pmatrix}$$

$\therefore$ the $\text{depth}(X; P)$ is independent of the particular homogeneous representation of X and P .

$$\therefore \frac{w}{T \|m_3\|} = \frac{m_3^T (\tilde{X} - \tilde{C})}{\|m_3\|}$$

$$\therefore \text{depth}(X; P)$$

$$= \frac{\text{sign}(\det M) m_3^T (\tilde{X} - \tilde{C})}{\|m_3\|} \quad (\text{see 16})$$

$$= \frac{\text{sign}(\det M) w}{T \|m_3\|}$$

where $M = \begin{pmatrix} m_1^T \\ m_2^T \\ m_3^T \end{pmatrix}$, proved.

$$18.3 \quad \text{---}$$

$$\# \text{depth}(X; P)$$

$$= \frac{\text{sign}(\det M) (kw)}{(kT) \|m_3\|}$$

$$= \text{depth}(X; P)$$

$$\text{depth}(X; kP)$$

$$= \frac{\text{sign}(\det M) \cdot \text{sign}(k) (kw)}{T \|m_3\| \cdot |k|} = \text{depth}(X; P)$$

18. depth of points

$X = (X, Y, Z, T)^T$ 3D point,

$P = (M, p_4)$ is camera matrix for a finite camera

$$x = PX, \quad x = (x, y, 1)^T$$

$\Rightarrow$ the depth of the point X

in front of the principal plane is

$$\text{depth}(X; P) = \frac{\text{sign}(\det M) w}{T \|m_3\|}$$

proof:

$$w = P_3^T X = P_3^T (X - \tilde{C})$$

$$(\because PC = 0)$$

$$\text{---}$$

$$\text{---}$$

$$w = P_3^T X = T P_3^T (\tilde{X}) = T P_3^T ((\tilde{X}) - (\tilde{C}))$$

$$(\because PC = 0)$$

$$\therefore w = T m_3^T (\tilde{X} - \tilde{C})$$

4. The set of points in space mapping to so that points on the scene plane a line l via the camera matrix P is the have zero z -coordinate.

plane $P^T l$.

proof: $x = PX, x \in l$

$$\Leftrightarrow x^T l = 0, x = PX$$

$$\Leftrightarrow X^T P^T l = 0, x = PX$$

$$\Leftrightarrow X^T \pi = 0, \pi = P^T l, x = PX$$

$$\Leftrightarrow X \in \pi, \pi = P^T l, x = PX$$

5. camera P , line l represented as a

~~Plicker~~ Plücker matrix, l is its image.

$$\Rightarrow l_x = PLP^T$$

proof: $a = PA, b = PB, a, b \in l, A, B \in L$.

let $M = PLP^T, \therefore M = P(AB^T - BA^T)P^T$

$$= a b^T - b a^T$$

$$= (a \times b)_x \quad (': \langle a \times b \rangle \text{ nullspace})$$

$$= l_x \quad (': l = a \times b)$$

5.4 The relation between l_i and l_{jk} is linear.

∴ This relation is quadratic in the elements of P .

Chapter 8 HZ

More Single View Geometry

1. Chapter 6 introduced the projection matrix as the model for the action of a camera on points.

This chapter describes the link between other 3D entities and their images under perspective projection. These entities include planes, lines, conics and quadrics.

2. The most general trans that can occur between a scene plane and an image plane under perspective imaging is a plane projective transformation. proof: (assume that the camera center does not lie on the scene plane).

Choose the world coordinate frame such that the XY -plane corresponds to a plane π in the scene,

$$\therefore x = PX$$

$$= \begin{pmatrix} p_1 & p_2 & p_3 & p_4 \end{pmatrix} \begin{pmatrix} X \\ Y \\ 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} p_1 & p_2 & p_3 & p_4 \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$= H x_u$$

where $x_u = \begin{pmatrix} X \\ Y \end{pmatrix}$

2.4. If the camera is affine, (i.e., $P_3 = 0$).

$\Rightarrow$ the scene and image planes are

related by an affine trans. $\therefore$

2.5. $P = K(R, t) \Rightarrow H = K(R, t)$,

where $R = (r_1, r_2, r_3)$.

A line in 3-space projects to a line in the image.

proof: $\therefore X(u) = A + uB$

$\therefore x(u) = P(X(u)) = PA + uPB$
 $= a + ub$

9.4 extended cross-product.

10. line L , $P_1 L = 0$

$\Rightarrow$ camera center $\in L$.

proof: $\because P_1 L = 0$

$$\therefore (P_2 \wedge P_3 | L) = 0 \quad \therefore P_2 \wedge P_3, L \text{ intersect.}$$

Similarly, $P_3 \wedge P_1, L$ intersect, $P_1 \wedge P_2, L$ intersect.

$\therefore$ camera center $= P_1 \cap P_2 \cap P_3$

$\therefore$ center $\in L$. (ε) .

11. A cone is a degenerate quadric.

12. Under the camera P the conic C

back-projects to the cone $Q_{\infty} = P^T C P$

proof: $x \in C \Leftrightarrow x^T C x = 0$

$$\Leftrightarrow x^T P^T C P x = 0$$

$$\Leftrightarrow x^T Q_{\infty} x = 0, Q_{\infty} = P^T C P$$

$$\Leftrightarrow x \in Q_{\infty}, Q_{\infty} = P^T C P$$

12.4 $Q_{\infty} C = P^T C P C = 0 \Rightarrow C$ is the vertex of the cone. ε

here C is camera center.

$$12.5 \quad P = K(I, 0) \Rightarrow Q_{\infty} = \begin{pmatrix} K^T C K & 0 \\ 0 & 0 \end{pmatrix}$$

proof:

$$L = axb = (PA) \times (PB)$$

$$\therefore L = \begin{vmatrix} P_2^T A & P_3^T B \\ P_3^T A & P_1^T B \end{vmatrix}$$

$$= (P_2 \wedge P_3 | L) \quad (\varepsilon) \quad (3.14 \text{ on } P_3)$$

The other components follow in

a similar manner.

$$9. \quad a \times b := \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = a_3 b$$

$$i = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, j = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, k = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$10 \quad a \times b \times c := \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (-1)^{m+1} (a, b, c) \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, n=4$$

$$11 \quad a \times b \times c \times d := \begin{vmatrix} i & j & k & l & m \\ a_1 & a_2 & a_3 & a_4 & a_5 \\ b_1 & b_2 & b_3 & b_4 & b_5 \\ c_1 & c_2 & c_3 & c_4 & c_5 \\ d_1 & d_2 & d_3 & d_4 & d_5 \end{vmatrix}$$

$$= (-1)^{m+1} (a, b, c, d), n=5$$

6. the map between the Plücker line coordinates L and the image line coordinates is represented by a matrix.

7. line projection matrix P is defined:

$$P := \begin{pmatrix} P_2 \wedge P_3 \\ P_3 \wedge P_1 \\ P_1 \wedge P_2 \end{pmatrix}, \quad (\text{we use } P_L)$$

$$\text{where } P = \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \end{pmatrix},$$

$P_1 \wedge P_2$ the Plücker line coordinates of the intersection of planes P_1 and P_2 .

$$7.4 \quad \text{rank}(P_L) = 3. \quad \varepsilon$$

8. P_L line projection matrix,

L line represented by Plücker line coordinates

$\Rightarrow$ the image line $L = P_L L$ (not right strictly)

$$= \begin{pmatrix} P_2 \wedge P_3 | L \\ P_3 \wedge P_1 | L \\ P_1 \wedge P_2 | L \end{pmatrix}$$

where $(L|L)$ is defined in 4.1 in 3.

proof: $\because x \rightarrow Hx$

$\because C \rightarrow H^T C H^{-1}$, C is conic

$\because C = \Omega_{\infty} = I$ on π_{∞} , $\therefore H = KR$

$\therefore IAC \ w = (KR)^T I (KR)^{-1}$

$$= (KK^T)^{-1}$$

17.4 Like Ω_{∞} the conic w is an

imaginary point conic with no real points.

17.5 conic w may be thought of as a convenient algebraic device.

18. the angle θ between two rays:

$$\cos \theta = \frac{x_1^T w x_2}{\sqrt{x_1^T w x_1} \sqrt{x_2^T w x_2}}$$

18.3. It is unchanged under projective trans of the image. $\therefore$

19. dual image of absolute conic (DIAC)

$$w^* = w^{-1} = KK^T, \quad \epsilon$$

$$19.3 \quad w^* = PG_{\infty}^* p, \quad \epsilon$$

19.4 $w \Rightarrow K$ (Cholesky factorization)

20 A point x and line L back-projecting to a ray and plane respectively that are orthogonal $\Rightarrow L = wx$

15. An image line L defines a plane

through the camera center with normal direction $n = K^T L$ measured in the camera's Euclidean coordinate frame. (Note: $\|n\| \neq 1$ in general).

proof: $\because d = K^T x, \quad d^T n = 0$

$$\therefore x^T K^T n = 0$$

$$\therefore L = K^T n$$

$$\therefore n = K^T L$$

16. the mapping between π_{∞} and an image is given by the planar homography

$$x = Hd \text{ with } H = KR.$$

proof: $\because X_{\infty} = \begin{pmatrix} d \\ 0 \end{pmatrix} \in \pi_{\infty}$

$$\therefore x = PX_{\infty} = KR \begin{pmatrix} I & -C \end{pmatrix} \begin{pmatrix} d \\ 0 \end{pmatrix}$$

$$= KRd$$

$$= Hd$$

the image of absolute conic (IAC) is the conic $w = (KK^T)^{-1} = K^{-T} K^{-1}$

13. The camera calibration matrix K is the (affine) trans between x and the ray's direction $d = K^T x$ measured in the camera's Euclidean coordinate frame.

proof: $\because \text{ray: } \tilde{X} = \lambda d$

$$\therefore x = K \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \lambda d \\ 1 \end{pmatrix} = Kd \text{ (up to scale)}$$

$$\therefore d = K^T x$$

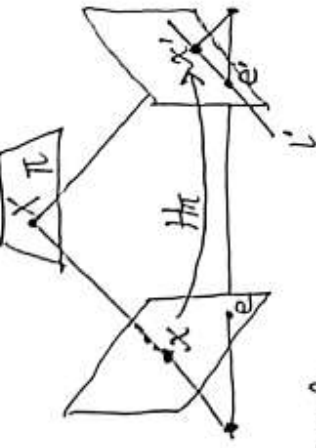
14. θ , the angle between two rays, with directions d_1, d_2 corresponding to image points x_1, x_2 respectively,

$$\Rightarrow \cos \theta = \frac{d_1^T d_2}{\|d_1\| \|d_2\|} = \frac{x_1^T M x_2}{\sqrt{x_1^T M x_1} \sqrt{x_2^T M x_2}} \quad \therefore$$

$$M = K^{-T} K^{-1}$$

14.3 $\because$ if K is known, then the angle between rays can be measured from their corresponding image points.

Chapter 7 HL



4. ~~$\exists$ some plane π~~

$$s.t. \{P^T x \mid x \in \text{image plane}\} = \pi$$

proof: $\forall x_1, x_2, x_3$ different, $\in \text{image plane}$

$$\Rightarrow x_1 = P^T x_1, x_2 = P^T x_2, x_3 = P^T x_3$$

x_1, x_2, x_3 different.

1st image plane $\xrightarrow{\text{projectively}} \pi$ (19 in C2)

$\therefore$ 1st image plane $\xrightarrow{\text{projectively}} \pi$ dense $\pi = \langle x_1, x_2, x_3 \rangle$

$\therefore \exists$ 2D homography H_π mapping $\forall x \in \text{image plane}$

each $x \in$ 1st image plane to $x' \in$ 2nd $\exists \lambda_1, \lambda_2, \lambda_3$ s.t. $x' = \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3$

$$\therefore l' = e' x x' = e' x H_\pi x$$

$$\therefore F = e' x H_\pi$$

5. $F = [e']_x H_\pi$, where H_π is the

Note: a plane π is not required in order transfer mapping from one image for F to exist. The plane π is simply to another via any plane π .

used as a means of defining a point $\text{rank}(F) = 2$.

map from one image to another. (here π not passing through either of the two camera centers)

Epipolar geometry and fundamental matrix
epipole, epipolar line, epipolar plane, epipolar pencil, baseline, converging cameras.

2. the fundamental matrix is the algebraic representation of epipolar geometry.

There is a map $x \rightarrow l'$ from a point in one image to its corresponding epipolar line in the other image.

3.4. This mapping is a singular correlation, which is represented by a matrix F , the fundamental matrix.

11. $P = K \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $P' = K(Rt)$

(here world origin at the first camera) $\Rightarrow t_x M = M^* (M^{-1}t)_x = M^{-T} (M^{-1}t)_x$
(up to scale)

$\Rightarrow P^+ = P^T (PP^T)^{-1} = \begin{pmatrix} K^{-1} \\ 0^T \end{pmatrix}$

$C = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$e = P \begin{pmatrix} -R^T t \\ 1 \end{pmatrix} = KR^T t$
(up to scale)

$e' = P' \begin{pmatrix} 0 \\ 1 \end{pmatrix} = K't$

$F = (P'C)_x P'P^+$

$= (K't)_x K'RK^{-1} = K'^{-T} t_x R K^{-1}$

$= K'^{-T} R (R^T t)_x K^{-1} = K'^{-T} R K^{-1} (KR^T t)_x$

$F = e'_x K' R K^{-1} = K'^{-T} R K^{-1} e_x$

12. $\forall$ pair of corresponding points $x \leftrightarrow x'$

in two images $\Rightarrow x'^T F x = 0$

12.4 If image points satisfy $x'^T F x = 0$

then the rays defined by these points are coplanar.

7. $\forall t$ vector, M invertible

$t_x M = M^* (M^{-1}t)_x = M^{-T} (M^{-1}t)_x$
(up to scale)

proof: let $x = M^{-1}t$, $\therefore t = Mx$

$t_x M = (Mx)_x M$

$M^* (M^{-1}t)_x = M^* x_x$

$M^{-T} (M^{-1}t)_x = M^{-T} x_x$

($\because M$ invertible $\Rightarrow M^* = \det(M) M^{-T}$)

8 $[a]_x = a_x a_x a_x$ (up to scale)

or $a_x^3 = -\|a\|^2 a_x$

9 $F = e'_x M$ is fundamental matrix ($\because C \neq C'$, $\therefore e' \neq 0$)

$\Rightarrow e'_x e'_x F = F$ (up to scale)

$\therefore F = e'_x M$, where $M = e'_x F$

10. $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $P' = (M t)$

$\Rightarrow F = t_x M$

(see 11)

5.5 given two cameras P, P'

camera centers C, C' , $C \neq C'$.

$\Rightarrow F = [e']_x P'P^+$

where $e' = P'C$, $PC=0$, $P'C'=0$

proof: $\therefore \{P^+x\}$, $x \in$ image plane $\therefore$ is a plane

$\therefore H_\pi = P'P^+$

$\therefore l' = e'_x (H_\pi x) = e'_x P'P^+ x$

$\therefore F = e'_x P'P^+$

here $e' = P'C$, C is solved from $PC=0$.

P^+ is pseudo-inverse of P . (17.3 in CG)

15. ~~F~~ F is fundamental matrix of the pair of cameras (P, P')

$\Rightarrow F^T$ is the fundamental matrix of the pair of cameras (P', P)

proof:

method 1: $\therefore F = e_x' H_\pi$

$$\begin{aligned} &= (H_\pi e_x)' H_\pi \\ &= H_\pi^* e_x \\ &= H_\pi^T e_x \\ &= (e_x' H_\pi^T)^T \\ &= (-e_x' H_\pi^T)^T \\ &= (e_x H_\pi^T)^T \end{aligned}$$

method 2:

$$\begin{aligned} \therefore x^T F x &= 0 \\ \Leftrightarrow x^T F^T x &= 0 \end{aligned}$$

from B, proved.

14. ~~the~~

given two cameras P, P' with non-coincident centers $\Rightarrow$ fundamental matrix exists and is unique.

proof: $\textcircled{1} F \exists$.

$\textcircled{2} \text{ If } \exists F_2 \neq F$.

$$\begin{aligned} \forall x_1, x_2, x_3, x_1 \neq x_2 \neq x_3 \\ \therefore l_1' = F x_1, l_2' = F x_2, l_3' = F x_3 \\ \lambda_1 l_1' = F_2 x_1, \lambda_2 l_2' = F_2 x_2, \lambda_3 l_3' = F_2 x_3 \end{aligned}$$

$$\begin{aligned} \textcircled{3} (\text{ If } l_i' \neq l_j', i \neq j \\ \Rightarrow l_i' + l_j' = F(x_i + x_j) \\ \Rightarrow l_i' + l_j' = F_2(x_i + x_j) = \lambda_i l_i' + \lambda_j l_j' \\ \therefore \lambda_i = \lambda_j = k \end{aligned}$$

$$\begin{aligned} k(l_i' + l_j') &= F_2(x_i + x_j) = \lambda_i l_i' + \lambda_j l_j' \\ \therefore \lambda_i &= \lambda_j = k \end{aligned}$$

$\textcircled{4} \text{ If } \text{rank}(l_1' l_2' l_3') = 2$

$\therefore \lambda_1 = \lambda_2 = \lambda_3$ (use $\textcircled{3}$)

$\textcircled{5} \text{ let } X = (x_1, x_2, x_3), L = (l_1, l_2, l_3) \therefore FX = L, F_2 X = \lambda L$
 $\therefore F = LX^{-1}, F_2 = \lambda LX^{-1}, \therefore F = F_2$ proved. **3**

13. two images acquired by cameras with non-coincident centers, $\Rightarrow$ fundamental matrix F is the unique 3×3 rank 2 homogeneous matrix which satisfies $x'^T F x = 0$ for all corresponding points $x \leftrightarrow x'$.

proof: $\textcircled{1} F \exists$.

$\textcircled{2} \text{ If } \exists F_2 \neq F$,

$\therefore x'^T F_2 x = 0, \forall x \leftrightarrow x'$

$\therefore F x = F_2 x, \forall x \in 1^{\text{st}} \text{ image plane}$

($\therefore$ if $\exists x, \text{ s.t. } l_1' = F x, l_2' = F_2 x, l_1' \neq l_2'$)

$\therefore \forall x' \in l_1', \Rightarrow x'^T l_1' = x'^T F x = x'^T F_2 x = x'^T l_2'$

$\therefore l_1' = l_2'$ error

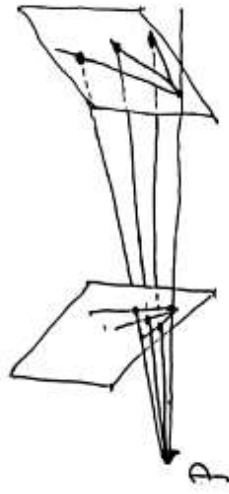
$\therefore F = F_2$ (from 14)

$\therefore$ error. proved.

21. The correspondence between epipolar lines, $l_i \leftrightarrow l'_i$, is defined by the pencil of planes with axis the baseline.

21.3 The corresponding lines are related by a perspectivity with center any point p on the baseline.

21.4 The correspondence between epipolar lines in the pencils is a 1D homography.



22. The def of the fundamental matrix can be counted as follows: 2 for e , 2 for e' , and 3 for the epipolar line homography which maps a line through e to a line through e' .

23. l, l' are corresponding epipolar lines, k is any line not passing through the epipole e , $\Rightarrow l' = F k_x l$, $l = F^T k'_x l'$

23.3 let $k = e$, $\Rightarrow l' = F e l$, $l = F^T e'_x l'$

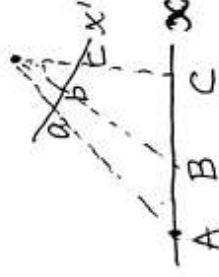
20. One special case of a projectivity is a perspectivity, the distinctive property

of which is that lines joining corresponding points

are concurrent. (24 in C2)

20.3 The composition of two perspectivities is a projectivity, but not, in general, a perspectivity.

20.4 1D perspectivity on the plane



20.5 2D perspectivity



16. $\det(F) = 7$

17. $ol' = Fx$ is the epipolar line corresponding to x .

18. $l = F^T x'$ is the epipolar line corresponding to x' .

19. $Fe = 0$, $F^T e' = 0$

20. Canonical cameras: (see 10)

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, P' = \begin{pmatrix} M & t \end{pmatrix}$$

$$\Rightarrow F = e'_x M = M^T e_x$$

$$\text{where } e' = t, e = M^{-1} t$$

21. central projection:

a ray from a point in space

is drawn from a 3D world point

through a fixed point in space,

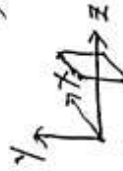
the center of projection.

The epipole in this case is termed the Focus of Expansion (FOE).



In the case of pure translation, ~~if the camera is moving~~

~~the camera~~, ~~the~~ points closer to the camera appear to move faster than those further away. — a common experience when looking out of a train window.



proof: ① $x = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$, $X = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$, $x = PX$

$\Rightarrow ZK^{-1}x = \begin{pmatrix} x \\ y \\ Z \end{pmatrix} = X$

② $\because Z = \text{depth}(X, P)$ (8 in C6)

③ $x' = P'X \Rightarrow x' = x + Kt/Z$ (inverse depth Z)
 proved.

6.11 If the world undergoes a translation $-t$, the camera is stationary (25),

$\Rightarrow$ points in 3d space move on straight lines parallel to t , and the ^{image} intersection of these parallel lines is the vanishing point v .

26.5 $v = e = e'$

$v = P \begin{pmatrix} t \\ 0 \end{pmatrix}$

$= Kt$

$\therefore v = e = e'$

26.6 $\therefore$ imaged parallel lines in 26.4 are the epipolar lines.

26.7 under the motion the epipole is a fixed point, i.e., has the same coordinates in both images, and points appear to move along lines radiating from the epipole

24. pure translation, pure planar motion the 'pure' indicates that there is no change in the internal parameters.

25. In considering pure translations of the camera, one may consider the equivalent situation in which the camera is stationary, and the world undergoes a translation $-t$.

26. $P = K(I, 0)$, $P' = K(I, -t)$ (pure translation)

$\Rightarrow F = e'x$ (see 11)

$e = e'$, $\text{def}(F) = 2$

$\Rightarrow x, x', e = e'$ are collinear if both images are overlaid on top of each other.
 ($\because x'x = x'^T F x = x'^T e'x = 0$
 $\therefore x \in l'$)

The collinearity property is termed auto-epipolar.

29.3 the first term takes account of

- the camera rotation and change of internal parameters; the second term takes account of camera translation.

30. geometric representation of the fundamental matrix.

31. The relationships $l' = Fx$ and $x'^T F x = 0$ are projective relationships. In other words they are projectively invariant.

31.2 $\hat{x} = Hx$, $\hat{x}' = H'x' \Rightarrow \hat{l} = \hat{F}\hat{x}$, with $\hat{F} = H'^T F H$, $\therefore H = K' R K^{-1}$

~~the~~ the corresponding fundamental matrix.

here H, H' projectivity.

32. H projective $\Rightarrow$ the fundamental matrices corresponding to the pairs of camera matrices (P, P') and (PH, PH) are same.

proof: $\because x = Px, x' = P'x$,
 $\therefore x = (PH)(H^{-1}x)$, $x' = (PH)(H^{-1}x)$
 x, x' still matched.
 from B, proved.

28. $P = K(I \ 0)$, $P' = K'(R \ 0)$

$$\Rightarrow F = e_x H, H = K' R K^{-1}$$

proof: $\because x = Px$

$$\hat{x} = \hat{P}x, \hat{P} = K'(R \ 0)$$

$$\therefore \hat{x} = K'(R \ 0)x$$

$$= K'R(I \ 0)x$$

$$= K'R K^{-1} K(I \ 0)x$$

$$= K'R K^{-1} P x$$

$$= K'R K^{-1} x$$

$$\therefore H = K'R K^{-1}$$

$$\textcircled{5} \therefore F = e_x H \quad (21)$$

29. $P = K(I \ 0)$, $P' = K'(R \ t)$

$$\Rightarrow x' = K' R K^{-1} x + K' t / z$$

where $x = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$, z from $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\text{proof: } \because x = \begin{pmatrix} z K^{-1} x \\ 1 \end{pmatrix},$$

$$\therefore x' = P' x = z K' R K^{-1} x + K' t$$

$$= K' R K^{-1} x + \frac{K' t}{z} \quad (\text{us 26.8})$$

27. General camera motion.

The first camera may be rotated and corrected to simulate a pure translational motion.

$$\Rightarrow F = e'_x H,$$

where F is the fundamental matrix for the original pair, e'_x is the fundamental matrix of the translation, and H is the projective transformation corresponding to the correction of the first camera.

proof: $\because \hat{F} = e'_x, x'^T \hat{F} \hat{x} = 0$, where $\hat{x} = Hx$ is the corrected point in the first image.

$$\therefore x'^T \hat{F} H x = 0, \therefore x'^T e'_x H x = 0$$

$$\therefore F = e'_x H.$$

35.

②. If $(P_1, \tilde{P}_1), (P_2, \tilde{P}_2)$

H_1, H_2 invertible, &

and $(P, \tilde{P}), (P', \tilde{P}')$ canonical,

$\therefore \exists \hat{H}$ invertible, s.t. $\tilde{P} = \hat{P}H, \tilde{P}' = P'H$.

$\therefore \exists H = H_1^{-1} \hat{H} H_2$ invertible,

$$(\tilde{P}H_2) = P'H_2 = (P'H_1)(H_1^{-1} \hat{H} H_2) = (P'H_1)H$$

$$(\tilde{P}'H_2) = P'\hat{H}H_2 = (P'H_1)(H_1^{-1} \hat{H} H_2) = (P'H_1)H$$

proved.

35.4. $\therefore$ the fundamental matrix captures

the projective relationship of two cameras.

35.5. A pair of camera matrices determines a

unique fundamental matrix.

② A given fundamental matrix determines the

pair of camera matrices up to right multiplication $\Rightarrow$

by a projective transformation.

35.6 $\text{def}(P, P') = 22$, $\text{def}(\text{projective world-frame}) = 15$

$$\text{def}(\text{fundamental matrix}) = 7, \quad 22 = 15 + 7$$

F fundamental matrix corresponding to each of the pairs $(P, P'), (\tilde{P}, \tilde{P}')$

$\Rightarrow \exists H$ invertible, s.t. $\tilde{P} = PH, \tilde{P}' = P'H$

proof: ① suppose each of the two pairs

is in canonical form.

$$\therefore P = \tilde{P} = (I \ 0)$$

$$P' = (A \ a), \tilde{P}' = (\tilde{A} \ \tilde{a})$$

$$\therefore F = a_x A = \tilde{a}_x \tilde{A} \quad (\text{see } |0|)$$

$$\therefore P' = (A \ a),$$

$$\tilde{P}' = (k^{-1}(A + av^T) \cdot ka)$$

(from 34)

$$\text{let } H = \begin{pmatrix} k^{-1}I & 0 \\ k^{-1}v^T & k \end{pmatrix}$$

$$PH = k^{-1}\tilde{P}, \quad P'H = \tilde{P}'$$

$(P, P'), (\tilde{P}, \tilde{P}')$ projectively related.

33. $(P, P') \Rightarrow \exists H$ invertible, s.t. $(PH, P'H)$ canonical form, i.e. $PH = (I \ 0)$

proof:

$$\text{let } P = \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \end{pmatrix}, \quad \therefore r(P) = 3,$$

$$\therefore \exists P_4, \text{ s.t. } \hat{P} = \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \\ P_4^T \end{pmatrix}, \quad r(\hat{P}) = 4.$$

$$\therefore \hat{P}\hat{P}^T = \begin{pmatrix} I & \\ & \end{pmatrix} \in 4 \times 4$$

$$\therefore P\hat{P}^T = \begin{pmatrix} I & 0 \end{pmatrix} \in 3 \times 4$$

$$\therefore H = \hat{P}^{-1}$$

34. matrix $F, r(F) = 2, F = \alpha_x A, F = \tilde{\alpha}_x \tilde{A}$

$$\Rightarrow \exists k \neq 0, v, \text{ s.t. } \tilde{\alpha} = k\alpha, \tilde{A} = k^{-1}(A + av^T)$$

$$\text{proof: } \textcircled{1} \therefore \tilde{\alpha}^T F = \tilde{\alpha}^T \alpha_x A = 0,$$

$$\tilde{\alpha}^T F = \tilde{\alpha}^T \tilde{\alpha}_x \tilde{A} = 0$$

$$r(F) = 2$$

$$\therefore \exists k \neq 0, \text{ s.t. } \tilde{\alpha} = k\alpha$$

$$\textcircled{2} \therefore \alpha_x A = \tilde{\alpha}_x \tilde{A} = k\alpha_x \tilde{A}$$

$$\therefore \alpha_x (k\tilde{A} - A) = 0, \therefore k\tilde{A} - A = \alpha v^T \text{ for some } v.$$

$$\therefore \tilde{A} = k^{-1}(A + \alpha v^T)$$

Camera matrices P, P'

if $r(P') = 3$, i.e., it is a valid camera,

$\Rightarrow P'FP$ skew-symmetric

$\Rightarrow F$ is the fundamental matrix

corresponding to the pair (P, P') . Proof: $P'FP$ skew-symmetric

$$\text{Proof: } (SF e)^T F (I 0) = \begin{pmatrix} F^T S^T F & 0 \\ e^T F & 0 \end{pmatrix}$$

which is skew-symmetric.

$\therefore$ proved. (from 38)

40.3 e' is epipole of P' .

$$\text{Proof: } PC = P' \begin{pmatrix} 0 \\ 1 \end{pmatrix} = e'$$

$$40.4 \quad r(SF) = 2.$$

$$\text{Proof: } \because r(S) = 2 \quad (27 \text{ in } C3)$$

$$r(F) = 2$$

$$\therefore r(SF) \leq 2.$$

$$\because r(SF e') = 3, \therefore r(SF) \geq 2. \therefore r(SF) = 2$$

$$40.5 \text{ If } S^T e' \neq 0, \text{ then } r(S^T F e') = 3$$

$$\text{Proof: } \because e^T F = 0, S^T e' \neq 0 \Rightarrow S^T \notin \perp F$$

$$\Rightarrow S^T F > 2 \quad (\text{from } 39) \Rightarrow e' \notin \perp SF \quad (\because S^T e' \neq 0)$$

41. The general formula for a pair of camera

matrices corresponding to a fundamental matrix $\therefore$ proved.

F is given by $P = (I \ 0)$,

$$P' = (e' F + e' e'^T \ \lambda e'), \text{ where } \lambda$$

any 3-vector, $\lambda \neq 0$. $\therefore$ (from 35, 40)

40 F matrix, $r(F) = 2$, S skew-symmetric,

$$P = \begin{pmatrix} I & 0 \end{pmatrix}, \quad P' = (SF \ e'), \quad e'^T F = 0, \quad 8$$

36. A real symmetric, $x^T A x = 0, \forall x,$

$$\Rightarrow A = 0$$

Proof: $\because A = UDU^T$, U orthogonal,

D real diagonal,

$$\therefore y^T D y = 0, \forall y.$$

$$\therefore D = 0 \quad \therefore A = 0$$

37. A skew-symmetric $\Leftrightarrow x^T A x = 0, \forall x.$

Proof: ① let $y = x^T A x$

$\Rightarrow$

$$\therefore 2y = y + y = x^T A x + x^T A^T x$$

$$\therefore 2y = x^T (A + A^T) x = 0 \quad \therefore y = 0$$

$$\textcircled{2} \Leftrightarrow x^T A x = 0, \forall x.$$

$$\therefore x^T B x = 0, \forall x, \quad B = A + A^T$$

$$\therefore B = 0 \quad (36)$$

$$\therefore A = -A^T$$

38. A non-zero matrix F is the fundamental matrix corresponding to a pair of

the square-roots of the eigenvalues of AA^T called essential matrix.

proof: $\tilde{A}\tilde{A} = V\tilde{D}^2V^T = V\tilde{D}^2V^T$ 43.3 $\therefore E = t_x R = R(R^T t_x)$

46.4 singular values are real and non-negative

47. matrix E is essential matrix

$\Leftrightarrow$ two of its singular values are equal, and the third is zero.

proof: E essential matrix $\Leftrightarrow E = t_x R$

$\Leftrightarrow E = SR$, S skew-symmetric

$\Leftrightarrow E = SR$, $S = kUZU^T$, U orthogonal, $Z = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

$\Leftrightarrow E = SR$, $S = U \text{diag}(1, 1, 0) W U^T$, $W = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$\Leftrightarrow E = U \text{diag}(1, 1, 0) V^T$, $V^T = W U^T R$

47.3 $E = (U \text{diag}(R_{xx}, 1) \text{diag}(1, 1, 0) (\text{diag}(R_{xx}, 1)) V^T$

R is any 2x2 rotation matrix.

48. the SVD of E is $U \text{diag}(1, 1, 0) V^T$, $E = SR$,

$\Rightarrow S = U Z U^T$, $R = U W V^T$ or $R = U W^T V^T$

48.3 $\therefore S \neq 0$, $\therefore t = U(\beta)$, $\therefore t = u_3$, $U = (u_1, u_2, u_3)$ SVD: $A = U D V^T$, $A: m \times n$,

49. $E = U \text{diag}(1, 1, 0) V^T$, $P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$,

$\Rightarrow P'$ has 4 possible choices.

$P' = (U W V^T u_3)$ or $P' = (U W V^T -u_3)$

or $P' = (U W^T V^T u_3)$ or $P' = (U W^T V^T -u_3)$

46.3. the singular values of A are

42. The essential matrix is the specialization of the fundamental matrix to the case of normalized image coordinates. The fundamental matrix may be thought of as the generalization of the essential matrix in which the (inessential) assumption of calibrated cameras is removed.

42.3 (calibration matrix K)

42.4 $P = K(R, t)$, $x = P X$,

① $\hat{x} = K^{-1} x$ is the image point expressed in normalized coordinates.

② $K^{-1} P = (R, t)$ is called a normalized camera matrix.

43. $P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $P' = (R, t)$,

the fundamental matrix corresponding to the pair of normalized cameras is

4.6. for calibrated cameras, the scene is determined at best up to a similarity trans.

proof: let $H_S = \begin{pmatrix} R & t \\ 0^T & \lambda \end{pmatrix}$ be any similarity trans.

$$P = K(R_p \ t_p)$$

$$\therefore PH_S^{-1} = K(R_p \ t_p) \begin{pmatrix} R^{-1} & -\lambda^{-1} R^{-1} t \\ 0^T & \lambda^{-1} \end{pmatrix}$$

$$= K(R_p R^{-1} \ t'),$$

$$\text{with } t' = -\lambda^{-1} R_p R^{-1} t + \lambda^{-1} t_p$$

$\therefore$ multiplying H_S^{-1} does not change the calibration matrix of P .

using 4.4, proved.

4.7 It is shown that for calibrated cameras, proj: $PX_i = (PH^{-1})(HX_i)$ reconstruction is possible up to a similarity trans. $P'X_i = (P'H^{-1})(HX_i)$

4.8 The reconstructed point set X_i and cameras differ from the true reconstruction by a trans belonging to a group (for instance a similarity). One speaks of projective reconstruction.

affine recon, similarity recon to indicate the type of ~~trans~~ trans involved.

cannot be determined from their images

are points on the baseline between two cameras. 3D reconstruction of cameras

4. ~~Without~~ reconstruction ambiguity. and structure.

4.3 Without some knowledge of a scene. The reconstruction task is to find the camera matrices P and P' , as well as the 3D points X_i such that $x_i = PX_i$, $x'_i = P'X_i$, $\forall i$.

2. outline of reconstruction method up to a Euclidean transformation (rotation and translation) with respect to the world frame.

4.4 the points X_i and the ^{uncalibrated} cameras can be determined at best only up to a projective (iii) $F \Rightarrow P, P'$

trans.

(iii) for each point correspondence $x_i \leftrightarrow x'_i$, compute the point X_i in space that projects to these two image points.

3. triangulation. 

4.5 It is shown that reconstruction from uncalibrated cameras is possible up to a projective trans.

3.3 The only points in 3-space that

7. the step to affine reconstruction

7.3 The essence of affine reconstruction

is to locate the plane at infinity

by some means, since this knowledge or $X_{2i}, HX_{1i} \in$ the line joining is equivalent to an affine reconstruction. the two camera centers.

7.4. The plane at infinity cannot be

identified unless some extra info is given.

7.5 projective reconstruction $(P, P', \{X_i\})$, 6. stratified reconstruction.

the plane at infinity π_{∞} in projective recon.

$$H = \begin{pmatrix} 1 & 0 \\ \pi_{\infty} \end{pmatrix}, \text{ if } \pi_{\infty}(4) \neq 0, \text{ i.e.}$$

$$H_0^{-1}$$

H_0 is Householder matrix

$$\text{s.t. } H_0 \pi_{\infty} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{if } \pi_{\infty}(4) = 0$$

$\Rightarrow (PH^{-1}, PH^{-1} \cdot \{HX_i\})$ is an affine

reconstruction.

proof: Use 5.3 in CB.

$$\textcircled{2} H^{-1} \pi_{\infty} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

③ Similarly, $HX_{1i}, X_{2i} \in$ same ray

through the camera center of P'_2 .

④ $\therefore X_{2i} = HX_{1i}$

or $X_{2i}, HX_{1i} \in$ the line joining the two camera centers.

$$\Leftrightarrow FX_i = 0, F^T X_i = 0$$

~~$X_{2i} = HX_{1i}$~~

or

5. $x_i \leftrightarrow x'_i$ correspondences between points in two images,

F-fundamental matrix uniquely determined by $x_i^T F x'_i = 0, \forall i$.

Let $(P_1, P'_1, \{X_i\})$ and $(P_2, P'_2, \{X_{2i}\})$ be two reconstructions of the $x_i \leftrightarrow x'_i$
 $\Rightarrow \exists H$ invertible such that

$$P_2 = P_1 H^{-1}, P'_2 = P'_1 H^{-1}, X_{2i} = H X_{1i}$$

$\forall i$, except for those i s.t. $F x_i = F^T x'_i = 0$

$\therefore F$ is the fundamental matrix

corresponding to (P_1, P'_1) and (P_2, P'_2) .

$\therefore \exists H$ invertible, s.t. $P_2 = P_1 H^{-1}, P'_2 = P'_1 H^{-1}$.

$$P_2 X_{2i} = X_{1i}$$

$$\therefore P_2 (H X_{1i}) = P_1 H^{-1} H X_{1i} = P_1 X_{1i} = X_{1i}$$

$\therefore H X_{1i}, X_{2i} \in$ same ray through

the camera center of P_2 .

or the camera calibration. 2

11. condition of 9

$\Rightarrow$ ~~affine reconstruction~~

$$F = t_x$$

$\Rightarrow$ affine reconstruction cameras

$$(P, P') \text{ with } P = (I \ 0), P' = (I \ t) \text{ proof: } \therefore P = (I \ 0), P' = (I \ t)$$

$$\therefore e = e' = t$$

12. $\{P_1, P'_1, \{X_{1i}\}\}$ and $\{P_2, P'_2, \{X_{2i}\}\}$

two reconstructions of $X_i \leftrightarrow X'_i$,

$$\exists H \text{ invertible, s.t. } P_2 = P_1 H^{-1}, P'_2 = P'_1 H^{-1}$$

$\Rightarrow$ the two reconstructions are up to a transformation H .

proof: use proof of 5.

13. $\exists (P, P')$ affine recon with $P = (I \ 0), P' = (M' \ t)$

$\Rightarrow$ the infinite homography $H_{\infty} = M'$.

(H_{∞} is unchanged by an affine trans of the cameras. $\hookrightarrow$)

Conversely, if H_{∞} has been obtained, then the cameras of an affine recon may be chosen

as $P = (I \ 0), P' = (H_{\infty} \ t)$. $\square$

9. the motion of the cameras is a pure translation with no rotation and no change in the internal parameters.

$\Rightarrow F$ is skew-symmetric.

$$\text{proof: } \therefore P = (I \ 0), P' = (I \ t)$$

$$\Rightarrow F = t_x$$

$\therefore F = t_x$ corresponding to $((M, m), (M, m+t))$ (from 8).

10. the motion of the cameras is a pure translation (condition 9)

$\Rightarrow$ for an affine reconstruction

one may choose the two cameras as $P = (I \ 0), P' = (I \ t)$.

proof: $\therefore$ true reconstruction

$$(M, m), (M, m+t)$$

$$\xrightarrow{\text{affine trans}} (P, P') \quad (\text{from 8})$$

(use 12)

(P, P') is the cameras of affine recon.

proof: $\therefore$ the present reconstruction

$\{HX_i\}$ differs from the true reconstruction by a projective-trans that fixes the plane at infinity (ℓ) , (from 5)

$\therefore$ a projective trans fixing the plane at infinity is an affine trans. (53 in C3)

$\therefore$ the reconstruction differs by an affine transformation from true recon.

$\therefore$ It is an affine reconstruction.

$$8. P = (M \ m) \neq (M, m+t), M \text{ invertible,}$$

$$\Rightarrow \exists H = \begin{pmatrix} M & m \\ 0^T & 1 \end{pmatrix}, \text{ s.t. } PH^{-1} = (I \ 0),$$

$$P'H^{-1} = (I \ t)$$

proof: $\square$

18. Just as the key to affine recon is the identification of the plane at infinity, the key to metric recon is the identification of the absolute conic.

proof: ~~the camera~~

19. the image of the absolute conic is a property of the image itself.

19.3. It is unchanged by 3D transformations of any particular reconstruction.

20. w , the image of the absolute conic,

$P = (M \ m)$ in an affine reconstruction.

$\Rightarrow$ the affine recon may be transformed to a metric recon by applying H :

$$H = \begin{pmatrix} A^T & 1 \end{pmatrix}, \text{ where } A \text{ is obtained by}$$

Cholesky factorization from $AA^T = (M^T w M)^{-1}$

~~proof~~ 21. direct metric reconstruction using w .

22. direct method.

Compute homography H such that $H X_i = X_{Ei}$ from five or more ground control points X_{Ei} with known Euclidean positions. Then the metric reconstruction is

$$P_m = P H^T, \quad P'_m = P' H^T, \quad X_{mi} = H X_i$$

proof of 13 & 14.

14. reconstruction:

① $F \rightarrow (P, P')$ using 41 in C9.

② upgrade the projective reconstruction to affine arithmetic reconstruction.

③ suppose the form (P, P')

i) $\Rightarrow F$ form

ii) $\Rightarrow F \Rightarrow (P, P')$ with form specified

④ affine recon may be chosen as the cameras of

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad P' = \begin{pmatrix} H_{00} & t \end{pmatrix}$$

i.e., we can use the form and solve it, then get an affine recon.

17. $(P, P', \{X_i\})$ projective recon for which

$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. In truth P is known to be projective.

an affine camera. $\Rightarrow$ an affine recon

is obtained by swapping the last two columns of P and P' and the last two

coordinates of each X_i . ($\because H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$)

15. we can assume projective recon cameras of

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad P' = \begin{pmatrix} M' & m' \end{pmatrix}.$$

⑤ The cameras of an affine recon

may be chosen as $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, P' = \begin{pmatrix} M' & m' \end{pmatrix}$ (finite camera required for 1st camera)

$$\Phi: \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} P \\ P_4 \end{pmatrix} = \begin{pmatrix} M & m \end{pmatrix} = P$$

$$\hat{P} = \begin{pmatrix} P \\ P_4 \end{pmatrix}, \quad P_4 \text{ can be chosen such that}$$

$$r(\hat{P}) = 4.$$

$$\textcircled{2} \because \hat{P} \text{ can be } \begin{pmatrix} M & m \\ 0 & 1 \end{pmatrix}, \quad r(\hat{P}) = 4$$

$\therefore$ Jacobian = I

if $x=y=0$.

2. $H = ART$, where

T is a translation, $Tx_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$,

R is a rotation about the origin, $R\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$

A is the mapping in 1.

$\Rightarrow H e = \begin{pmatrix} f \\ g \end{pmatrix}$, H is to first-order

a rigid transformation in the neighbourhood of x_0

3. (H, H') are called a matched pair of

transformations if $H^T L = H'^T L'$, $\therefore$ Jacobian = $\frac{\partial(x, y)}{\partial(x, y)}$

A pair of corresponding epipolar lines l, l' .

$$\therefore \frac{1}{1-t} = 1+t+t^2+\dots, |t| < 1$$

$$\therefore \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1-\frac{x}{f} \end{pmatrix}$$

$$\frac{1}{1-\frac{x}{f}} x$$

$$\begin{pmatrix} 1 \\ \frac{1}{1-\frac{x}{f}} y \end{pmatrix}$$

Chapter 11. compute P .

$$1. G = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1/f & 0 & 1 \end{pmatrix}, \quad \text{epipolar} = (f, 0, 1)^T$$

$$\Rightarrow {}^0 G e = \begin{pmatrix} f \\ g \end{pmatrix}$$

② G is approximated (to first-order)

at the origin by the identity mapping

$$\text{proof: } {}^0 G e = \begin{pmatrix} 1 & 0 & 0 \\ -1/f & 0 & 1 \end{pmatrix} \begin{pmatrix} f \\ g \\ 1 \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix}$$

$$\textcircled{2} \mathcal{H}^* \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = A x = A \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix}$$

$$\therefore \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = A \begin{pmatrix} \hat{x} \\ \hat{y} \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 1 - \frac{x}{f} \end{pmatrix}$$

Z is higher order terms in x and y .

6. ~~H, H'~~

outline of resampling algorithm:

input: a pair of images.

output: a pair of images resampled

so that the epipolar lines in the H, H' matched.

two images are horizontal, H, H' matched. $H = H_0 H_1$ is 5.

and such that corresponding points H, H' are as close to each other as then H' becomes

possible. Any remaining disparity $\min \sum d(H \hat{x}_i, \hat{x}'_i)^2$ ①

between matching points will be $H_0 \hat{x}_i$ where $\hat{x}'_i = H_0 \hat{x}_i$, $\hat{x}'_i = H' \hat{x}_i$

along the horizontal epipolar lines. 7.3 let $\hat{x}_i = (\hat{x}_i, \hat{y}_i, 1)^T$

(i) identify a seed set of matching points $\hat{x}_i \leftrightarrow \hat{x}'_i$ ($n \geq 7$) $\hat{x}'_i = (\hat{x}'_i, \hat{y}'_i, 1)^T$

(ii) compute F , find e, e' ① becomes

(iii) select H' , s.t. $H'e' = (b)$ using 2. $\min \sum (a\hat{x}_i + b\hat{y}_i + c - \hat{x}'_i)^2 + (\hat{y}_i - \hat{y}'_i)^2$ where $H_0 = H'M$,

(iv) find H , s.t. $\sum d(H\hat{x}_i, H'e')^2$ is minimized. using 7.4 $\min \sum (a\hat{x}_i + b\hat{y}_i + c - \hat{x}'_i)^2$ 2.

4. J, J' images, $F = e'M$,

H' a projective trans of J' ,

H a projective trans of J

$\Rightarrow H$ matches H'

$\Leftrightarrow H = (I + H'e'a^T)H'M$ for some a .

5. J, J' images, $F = e'M$,

H' a projective trans of J' ,

H a projective trans of J ,

$H'e' = (b)$,

$\Rightarrow H$ matches H'

$\Leftrightarrow H = H_0 H_1$,

where $H_0 = H'M$,

$H_1 = \begin{pmatrix} a & b & c \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

7. $\because x = PX \Leftrightarrow x \times (PX) = 0$

$\therefore$ we get three equations for each image ~~point~~ point, of which two are linearly independent. $\therefore$

7.3. $\therefore AX = 0$

with $A = \begin{pmatrix} x_1^T P_1^T - P_1^T \\ y_1^T P_1^T - P_1^T \\ x_2^T P_1^T - P_1^T \\ y_2^T P_1^T - P_1^T \end{pmatrix}$

where $x = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$, $x' = \begin{pmatrix} x' \\ y' \\ 1 \end{pmatrix}$, $P = \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \end{pmatrix}$

$P' = \begin{pmatrix} P_1^T \\ P_2^T \\ P_3^T \end{pmatrix}$

8. Homogeneous method (DLT) and inhomogeneous method.

8.3 They are not projective-invariant.

4. two rays corresponding to a matching pair of points $x \leftrightarrow x'$ will meet in space

$\Leftrightarrow$ the two points satisfy the epipolar constraint: $x'^T F x = 0$.

5. A triangulation is said to be invariant under a transform H

if $\tau(x, x', P, P') = H^{-1} \tau(x, x', HP, HP')$

where τ is a triangulation method used to compute X .

6. The estimated point in linear triangulation methods does not exactly satisfy the geometric relations, and is not an optimal estimate.

Chapter 12 HZ

Structure Computation

1. compute the position of a point in 3-space given its images in two views and the camera matrices of these views.

1.3 It is assumed that there are errors not in the P, P'

2. There is no meaningful metric info about the object space in affine and projective reconstruction.

3. ΦF is given a priori and then X is determined.

② F and $\{X_i\}$ are estimated simultaneously.

12. $\min_t C = d(x, l(t))^2 + d(x, l'(t))^2$ 9.14 Once $\hat{x}'$ and $\hat{x}$ are found, the point

12.3 $12 \Leftrightarrow 9$ $\hat{x}$ may be found by any triangulation

12.4 (l, l') are a pair of matching meet precisely in space.

epipolar lines parametrized by t . 10. real inner space $\mathbb{R}^n$

12.5 the optimum point $\hat{x}$ is the closest point on the line l to the point x and $\hat{x}'$ is similarly defined.

12.6 strategy for minimizing 12.

(i) write l in the first image as $l(t)$. (first-order geometric correction)

(ii) using F , compute $l'(t)$ in the second image. The approximation is accurate if the correction in each image is small (less than a pixel), and is cheap to compute.

(iii) express C as a function of t .

(iv) Find t .

However, the corrected points will not satisfy the epipolar relation

$$\hat{x}'^T F \hat{x} = 0 \text{ exactly.}$$

8.4 the inhomogeneous method is affine-invariant, whereas the homogeneous method is not.

8.5 The homogeneous linear method provides acceptable results.

It has the virtue that it generalizes easily to triangulation when more than two views of the point are available.

9. geometric error cost function.

$$\min_{\hat{x}, \hat{x}'} C(x, x') = \|\hat{x} - \hat{x}'\|^2 + \|\hat{x}' - \hat{x}\|^2$$

$$\text{s.t. } \hat{x}'^T F \hat{x} = 0$$

9.3 assuming a Gaussian error distribution,

$\Rightarrow \hat{x}', \hat{x}$ are MLE for the true image point correspondences.